

WEEK 10: PAPER 1 REVIEW — DATA TRANSMISSION & HARDWARE

LESSON 10.1 — Data Transmission, Error Detection & Encryption

Topic: Packet switching, transmission modes, error detection, encryption

Duration: 40 minutes

Lesson Type: Consolidation / Exam Practice

Syllabus Ref: 1.1.1–1.1.6 (CIE 0478)

Learning Objective:

By the end of this lesson, students will be able to explain packet switching, compare transmission modes, describe error detection methods, and evaluate encryption types in exam-style responses.

Key Vocabulary:

packet switching — 分组交换 fēn zǔ jiāo huàn

packet header — 分组头部 fēn zǔ tóu bù

router — 路由器 lù yóu qì

serial — 串行 chuàn xíng

parallel — 并行 bìng xíng

simplex — 单工 dān gōng

half-duplex — 半双工 bàn shuāng gōng

full-duplex — 全双工 quán shuāng gōng

parity check — 奇偶校验 qí ǒu jiào yàn

checksum — 校验和 jiào yàn hé

encryption — 加密 jiā mì

symmetric — 对称的 duì chèn de

asymmetric — 非对称的 fēi duì chèn de

SSL / TLS — 安全传输层协议 ān quán chuán shū céng xié yì

private key — 私钥 sī yuè

public key — 公钥 gōng yuè

Lesson Structure (40 Minutes):

0–3 min | DO NOW

Display on board: "A 5 MB image is sent over the Internet from Shenzhen to London.

List three things that happen to the data during transmission." (3 things)

Students write in books. Collect 2–3 responses. Key point: data is broken into packets, each may take different routes, reassembled at destination.

3–12 min | DIRECT INSTRUCTION

Topic 1: PACKET SWITCHING (4 min)

- Data split into packets (header + payload + footer/trailer)
- Header contains: source IP, destination IP, packet number, total packets
- Packets travel independently via routers → may take different paths
- Reassembled at destination using sequence numbers
- Advantages: fault tolerance (one route fails, others work), efficient use of bandwidth,

no dedicated line needed

- Disadvantage: packets may arrive out of order; overhead from headers

Draw diagram: Sender → [packets 1,2,3] → different routes through routers →

Reassembler → Receiver

Topic 2: TRANSMISSION MODES (3 min)

Create comparison table on board:

Mode	Direction	Example
Simplex	One way only	TV broadcast, radio
Half-duplex	Both ways, not simul.	Walkie-talkie
Full-duplex	Both ways simultaneously	Phone call
Serial	One bit at a time	USB, Ethernet cable
Parallel	Multiple bits at once	Old printer port

Key exam point: USB is SERIAL transmission.

Topic 3: ERROR DETECTION (2 min)

- Parity check: even/odd parity bit added; receiver checks if count matches.

Limitation: only detects odd number of bit errors; cannot correct.

- Checksum: sender calculates sum of data, sends with data; receiver recalculates and compares. More reliable than parity.
- Echo check: data sent back to sender for verification (inefficient).

Topic 4: ENCRYPTION (3 min)

- Symmetric: same key for encryption and decryption. Fast but key distribution problem.
- Asymmetric: public key encrypts, private key decrypts. Solved distribution problem.

Used in SSL/TLS for HTTPS.

- SSL/TLS: handshake process establishes secure connection; uses both symmetric and

asymmetric encryption during session.

12–25 min | GUIDED PRACTICE

Worked exam question (on board + student whiteboards):

"Explain the process of packet switching and give two advantages over circuit switching." (6 marks)

Step through mark scheme together:

- Data is split into packets (1)
- Each packet has a header with source/destination IP and sequence number (1)
- Packets travel independently via routers (1)
- Packets may take different routes (1)
- Reassembled at destination using sequence numbers (1)
- Advantages: no dedicated line needed / fault tolerance / efficient bandwidth (1 each, max 2)

Second worked question: "Compare symmetric and asymmetric encryption." (4 marks)

Students pair-share responses, then class check against mark scheme.

25–35 min | INDEPENDENT PRACTICE

Students complete WORKSHEET 10A (Q1–Q5). Teacher circulates, provides 1-to-1 support.

Extension for Secure students: begin Q6–Q8 (stretch questions on SSL handshake).

35–40 min | PLENARY + HOMEWORK

Plenary: "One thing you remembered, one thing you learned, one thing you're unsure about"

(3-2-1 reflection — write in books).

Homework: Complete WORKSHEET 10A. Read CIE syllabus sections 1.1.1–1.1.6.

EAL Support:

- Bilingual keyword list provided at start of lesson (pinned to desk)
- Sentence starters for exam answers: "Packet switching works by..."
- Visual diagram of packet structure with labels in both languages
- Partner EAL students with stronger English speakers during pair-share

Differentiation:

Emerging: Provide completed diagram of packet switching; focus on 2-mark questions; sentence starters provided; work with teacher on Q1–Q3

Developing: Complete standard worksheet Q1–Q6 independently; attempt 4-mark explain questions

Secure: Complete all questions including stretch Q7–Q8; peer-teach emerging students; attempt 6-mark extended questions with full technical vocabulary

Resources Needed:

- Projector + board
- Student whiteboards and markers
- Printed bilingual keyword sheets
- WORKSHEET 10A (printed)
- Timer (visible)

--- LESSON 10.2 ---

LESSON 10.2 — CPU, Memory, Storage & I/O Devices

Topic: CPU architecture, FDE cycle, memory hierarchy, storage, I/O, embedded systems

Duration: 40 minutes

Lesson Type: Consolidation / Exam Practice

Syllabus Ref: 1.2, 1.3, 1.4, 1.5 (CIE 0478)

Learning Objective:

By the end of this lesson, students will be able to describe the Von Neumann architecture, explain the FDE cycle with register-level detail, compare storage types, and identify I/O devices and embedded systems.

Key Vocabulary:

Von Neumann architecture — 冯·诺依曼架构 féng nuò yī màn jià gòu

Fetch-Decode-Execute — 取指-译码-执行 qǔ zhǐ - yì mǎ - zhí xíng

Program Counter (PC) — 程序计数器 chéng xù jì shù qì

Memory Address Register — 存储器地址寄存器 cún chǔ qì dì zhǐ jì cún qì

Memory Data Register — 存储器数据寄存器 cún chǔ qì shù jù jì cún qì

Accumulator (ACC) — 累加器 lěi jiā qì

Current Instruction Register — 当前指令寄存器 dāng qián zhǐ lìng jì cún qì

RAM — 随机存取存储器 suí jī cún qǔ cún chǔ qì

ROM — 只读存储器 zhǐ dú cún chǔ qì

cache — 高速缓存 gāo sù huǎn cún

magnetic storage — 磁存储 cí cún chǔ

optical storage — 光存储 guāng cún chǔ

solid-state — 固态 gù tài

embedded system — 嵌入式系统 qiàn rù shì xì tǒng

sensor — 传感器 chuán gǎn qì

actuator — 执行器 zhí xíng qì

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"Name the five registers involved in the FDE cycle and state the role of each."

Students write from memory. Check against correct answers after 2 minutes.

3–12 min | DIRECT INSTRUCTION

Rapid review of FOUR major hardware topics:

1. CPU & VON NEUMANN (3 min)

- Five key registers:
 - PC: holds address of NEXT instruction to fetch

- MAR: holds address of memory location being accessed
- MDR: holds data/instruction being transferred to/from memory
- CIR: holds current instruction being decoded
- ACC: stores results of arithmetic/logic operations
- Three buses: address bus (carries addresses, one-way), data bus (carries data, two-way), control bus (carries signals, two-way)
- FDE cycle: PC → MAR → memory → MDR → CIR → decoded → executed → PC increments
- Clock speed, cores, cache size affect performance
- Cache: small, fast memory between CPU and RAM; stores frequently used data

2. MEMORY (2 min)

Feature	RAM	ROM	
Volatile?	Yes (lost when off)	No (permanent)	
Writable?	Read and write	Read only	
Use	Store running programs	Store boot instructions	
Type	DRAM/SRAM	PROM/EPROM/EEPROM	

Memory hierarchy: Registers → Cache → RAM → Secondary Storage (speed vs capacity)

3. STORAGE (2 min)

Type	Technology	Example	Advantage	Disadvantage	
Magnetic	Coating/platter	HDD	Cheap, high cap.	Slow, fragile	
Optical	Laser reads pits	CD/DVD/Blu-ray	Portable	Lower capacity	
Solid-St.	Flash memory	SSD/USB	Fast, durable	Expensive/GB	

4. I/O & EMBEDDED SYSTEMS (2 min)

- Input devices: keyboard, mouse, scanner, microphone, barcode reader, sensors
- Output devices: monitor, printer, speaker, actuator, projector
- Sensors: temperature, pressure, light, motion, pH, gas, humidity, sound
- Actuators: motor, pump, valve, heater, buzzer, light (convert electrical → physical)
- Embedded system: dedicated computer inside a larger device; has processor, memory,

I/O; examples: washing machine, microwave, car ECU, digital camera

- Monitoring: measures and displays (no action). Control: measures AND takes action.

12–25 min | GUIDED PRACTICE

"Trace the FDE cycle for the instruction ADD 5" — whole class on board:

5. PC value (e.g., 100) copied to MAR
6. Address bus carries 100 to memory; control bus sends READ signal
7. Contents of address 100 (the instruction ADD 5) put on data bus → MDR
8. MDR copies instruction to CIR
9. PC increments (101)
10. Instruction decoded: ADD operation, operand is 5

11. ALU executes: $ACC + 5 \rightarrow$ result in ACC

Then: "List three embedded systems you used this morning and state their purpose."

25–35 min | INDEPENDENT PRACTICE

Students complete WORKSHEET 10B. Teacher circulates with questioning to probe understanding.

Focus questions: Q4 (FDE trace), Q7 (storage comparison), Q9 (embedded systems).

35–40 min | PLENARY + HOMEWORK

Plenary: Quick-fire quiz — 10 questions, 20 seconds each:

12. What does the PC store? (Address of next instruction)
13. What bus carries memory addresses? (Address bus)
14. Is cache faster or slower than RAM? (Faster)
15. Name a solid-state storage device. (SSD/USB)
16. What does an actuator do? (Converts electrical to physical action)
17. Is a washing machine an embedded system? (Yes)
18. What is the difference between monitoring and control?
19. Is ROM volatile? (No)
20. What does EEPROM stand for?
21. How many directions can data travel on the data bus? (Two)

Homework: Complete WORKSHEET 10B. Create a mind map of all CPU/memory/storage content.

EAL Support:

- Pre-printed FDE cycle diagram with blanks to fill in
- Bilingual glossary pinned to desk for all hardware terms
- Visual memory hierarchy diagram with sizes/speeds labelled
- Sentence frames: "The [register] is used to [function]."
- Allow diagram-based answers where language is a barrier

Differentiation:

Emerging: Provided with completed FDE cycle flowchart; matching exercise for registers and functions; supported by TA; focus on 1-2 mark recall questions

Developing: Complete worksheet independently; trace FDE with given instructions; compare two storage types in 2–3 sentences

Secure: Trace FDE with complex instructions (e.g., LDA, ADD, STA sequence); evaluate suitability of storage for given scenarios; explain why cache improves performance using technical vocabulary; peer-teach

Resources Needed:

- Projector
- Pre-printed FDE cycle diagram (with blanks)
- Printed bilingual keyword sheets
- WORKSHEET 10B (printed)
- Quick-fire quiz questions (on slide)

--- WORKSHEET 10A ---

WORKSHEET 10A: Data Transmission, Error Detection & Encryption

Student Name: _____ Date: _____

SECTION A: SHORT ANSWER (2 marks each)

Q1. Define packet switching. [2]

Q2. State two pieces of information stored in a packet header. [2]

Q3. Explain the difference between serial and parallel data transmission. [2]

Q4. Give one example of:

a) Simplex transmission: _____ [1]

b) Half-duplex transmission: _____ [1]

c) Full-duplex transmission: _____ [1]

Q5. Describe how a parity check works. [2]

SECTION B: EXAM-STYLE QUESTIONS

Q6. Explain the difference between symmetric and asymmetric encryption. [4]

Q7. Describe the process of packet switching. In your answer, explain how packets are created, routed, and reassembled. [6]

Q8. A student is buying a product from an online shop. The website uses HTTPS.

a) Explain how SSL/TLS keeps the transaction secure. [3]

b) State one advantage of using a checksum over a parity check for error detection. [1]

Q9. A company needs to choose between serial and parallel transmission for a new printer connection. The printer is 3 metres from the computer.

Evaluate which transmission method would be more suitable. [4]

--- WORKSHEET 10B ---

WORKSHEET 10B: CPU, Memory, Storage & I/O Devices

Student Name: _____ Date: _____

SECTION A: RECALL QUESTIONS

Q1. Name the FIVE registers in the Von Neumann architecture and state the function of each. [5]

Q2. a) State the three buses in the Von Neumann architecture. [3]

b) For each bus, state what it carries and its direction. [3]

Q3. Complete the table comparing RAM and ROM: [4]

Feature	RAM	ROM
Volatile?		
Writable?		
Purpose		

SECTION B: EXAM-STYLE QUESTIONS

Q4. Trace the Fetch-Decode-Execute cycle for the instruction "ADD 7".

Your answer should include the role of each register. [6]

Q5. Explain how cache memory improves CPU performance. [2]

Q6. A photographer needs to store 50,000 high-resolution images (total 2 TB).

Evaluate which storage type would be most suitable. Consider magnetic, optical, and solid-state options. [4]

Q7. Describe the difference between a sensor and an actuator.

Give one example of each. [4]

Q8. A washing machine is described as an embedded system.

- a) Define an embedded system. [2]
- b) Give two other examples of embedded systems. [2]
- c) Explain the difference between monitoring and control in an automated system. [2]

Q9. Describe the memory hierarchy, explaining the trade-off between speed and capacity at each level. [4]

Q10. State three factors that affect CPU performance and explain how each affects the speed of processing. [6]

--- ANSWER KEY (WEEK 10) ---

WORKSHEET 10A — ANSWER KEY

Q1. Packet switching is the method where data is split into packets (1) that travel independently across a network and are reassembled at the destination (1).

Q2. Any two from: source IP address, destination IP address, packet number/sequence number, total number of packets, error-checking data (1 each, max 2).

Q3. Serial transmission sends data one bit at a time (1) along a single channel; parallel transmission sends multiple bits simultaneously (1) along multiple channels.

Q4. a) Any one: TV broadcast, radio, keyboard to computer (1)

b) Any one: Walkie-talkie, older CB radio (1)

c) Any one: Telephone/mobile phone call, video chat (1)

Q5. A parity check adds an extra parity bit to make the total number of 1s either even or odd (1); the receiver counts the 1s and checks against the agreed parity (1).
If the count doesn't match, an error is detected.

Q6. • Symmetric encryption uses the same key to encrypt and decrypt data (1)

- It is faster but has the key distribution problem (1)
- Asymmetric encryption uses a public key to encrypt and a private key to decrypt (1)

- It solves the key distribution problem but is slower (1)

Q7. Mark scheme (6 marks):

- Data is split into small packets (1)
- Each packet contains a header with source/destination IP and sequence number (1)
- Packets are sent independently across the network (1)
- Packets may take different routes via routers (1)
- Packets are reassembled at the destination in the correct order using sequence numbers (1)

- Any advantage stated: fault tolerance, efficient use of bandwidth, no dedicated line needed (1)

Q8. a) • SSL/TLS uses asymmetric encryption during the handshake to establish a secure connection (1)

- A session key is then created for symmetric encryption (1)
- This means data sent between browser and server is encrypted and cannot be read by interceptors (1)

b) A checksum can detect multiple bit errors / is more reliable than parity (1)

Q9. For a 3-metre connection:

- Serial would be more suitable (1)
- Parallel suffers from skew/interference over longer distances (1)
- Serial is more reliable over longer distances / modern serial (USB) is fast enough for printers (1)
- Parallel requires more wires so cables are thicker and more expensive (1)

WORKSHEET 10B — ANSWER KEY

Q1. (1 mark each, max 5)

- PC (Program Counter): stores address of the next instruction to be fetched
- MAR (Memory Address Register): stores the address of the memory location being read from or written to
- MDR (Memory Data Register): stores the data/instruction being transferred to or from memory
- CIR (Current Instruction Register): stores the current instruction being decoded
- ACC (Accumulator): stores the result of arithmetic and logic operations

Q2. a) Address bus, Data bus, Control bus (1 each, max 3)

- b) • Address bus: carries memory addresses, one-way (1)
- Data bus: carries data/instructions, two-way (1)
 - Control bus: carries control signals (read/write/clock), two-way (1)

Q3. (1 mark per correct cell, max 4)

Feature	RAM	ROM	
Volatile?	Yes	No	

Writable?	Yes	No (mostly)	
Purpose	Store OS,	Store boot/	
	programs	startup	
	currently	instructions	
	running		

Q4. Mark scheme (6 marks):

- 22. The PC holds the address of the instruction ADD 7 (1)
- 23. This address is copied to the MAR (1)
- 24. The address bus carries the address to memory; control bus sends READ signal (1)
- 25. The instruction ADD 7 is placed on the data bus and copied to the MDR (1)
- 26. The instruction is copied from MDR to the CIR (1)
- 27. The PC is incremented (1)
- 28. The instruction is decoded: the operation is ADD and the operand is 7 (1)
- 29. The ALU adds 7 to the value in the ACC; result stored in ACC (1)

(Max 6 — award best 6 points)

Q5. Cache is a small, fast memory located between the CPU and RAM (1).

It stores frequently accessed data so the CPU does not need to wait for slower RAM access, reducing memory access time (1).

Q6. Magnetic (HDD) would be most suitable (1):

- Much cheaper per TB than SSD (1)
- 2TB capacity easily available (1)
- Optical would require many discs, impractical for 2TB (1)
- SSD would be very expensive for this capacity (1)

(Accept solid-state if well justified — e.g., speed needed for professional photography workflow. Award max 3 if not compared against alternatives.)

Q7. A sensor detects/measures physical conditions and converts them to electrical signals (1).

Example: temperature sensor, light sensor, pressure sensor (1).

An actuator converts electrical signals into physical movement/action (1).

Example: motor, pump, buzzer, heater, valve (1).

Q8. a) An embedded system is a dedicated computer system built into a larger device (1) to perform a specific task (1).

b) Any two: microwave oven, digital camera, car engine management system, smart thermostat, fitness tracker, dishwasher, ATM (1 each)

c) Monitoring measures/display data without taking action (1).

Control measures data AND takes action based on the readings (1).

Q9. Registers (fastest, smallest) → Cache → RAM → Secondary Storage (slowest, largest) (1)

As you move down the hierarchy, capacity increases but speed decreases (1).

This is because faster memory technologies are more expensive per bit (1).

The hierarchy gives the best balance of speed and cost (1).

Q10. (2 marks each pair, max 6)

- Clock speed: higher clock speed means more cycles per second, more instructions processed per second (1+1)
- Number of cores: more cores means more instructions can be processed simultaneously (1+1)
- Cache size: larger cache means more frequently used data can be stored close to the CPU, reducing RAM access (1+1)

WEEK 11: PAPER 1 REVIEW — SOFTWARE & THE INTERNET

LESSON 11.1 — System Software, Languages & Translation

Topic: OS functions, software types, programming languages, translators, IDEs

Duration: 40 minutes

Lesson Type: Consolidation / Exam Practice

Syllabus Ref: 1.2, 2.1 (CIE 0478)

Learning Objective:

By the end of this lesson, students will be able to classify software types, explain OS functions and interrupts, compare translators, and describe IDE features.

Key Vocabulary:

system software	— 系统软件 xì tǒng ruǎn jiàn
application software	— 应用软件 yìng yòng ruǎn jiàn
operating system	— 操作系统 cāo zuò xì tǒng
interrupt	— 中断 zhōng duàn
firmware	— 固件 gù jiàn
high-level language	— 高级语言 gāo jí yǔ yán
low-level language	— 低级语言 dī jí yǔ yán
machine code	— 机器码 jī qì mǎ
compiler	— 编译器 biān yì qì
interpreter	— 解释器 jiě shì qì
assembler	— 汇编器 huì biān qì
IDE	— 集成开发环境 jí chéng kāi fā huán jìng
source code	— 源代码 yuán dài mǎ
object code	— 目标代码 mù biāo dài mǎ
library program	— 库程序 kù chéng xù
utility software	— 实用软件 shí yòng ruǎn jiàn

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"Sort these into system software and application software: Microsoft Word, Windows 11, Norton Antivirus, Photoshop, Linux kernel, Disk defragmenter, WeChat, Device driver." Write in books. Check: Word, Photoshop, WeChat = app; Windows, Linux, Norton, defragmenter, driver = system.

3–12 min | DIRECT INSTRUCTION

Topic 1: SOFTWARE TYPES (2 min)

- System software: manages computer hardware and provides platform for apps. Includes OS, utility software, device drivers, library programs.
- Application software: performs tasks for the user. General purpose (word processor, spreadsheet, database, browser, presentation software) vs specialist/bespoke (custom-written for specific organisation).
- Utility software: antivirus, disk defragmenter, disk formatter, backup software, file compression, firewall.

Topic 2: OS FUNCTIONS (4 min) — ALL NINE

30. Memory management: allocates RAM to programs, manages virtual memory, keeps track of used/free memory.
31. File management: organises storage, directories/folders, permissions.
32. Input/Output management: communicates with peripherals via device drivers.
33. Interrupt handling: receives and processes interrupt signals; priorities them.
34. Security: user accounts, passwords, access rights.
35. User interface: CLI vs GUI — provides way for user to interact.
36. Processor/multitasking scheduling: allocates CPU time to processes.
37. Error reporting: detects and reports hardware/software errors.
38. Loading and executing programs: transfers programs from storage to RAM and runs.
- INTERRUPTS: signal sent to CPU to gain attention. Sources: hardware error, user input, timer, I/O request. ISR (Interrupt Service Routine) handles it.
- Priority system: some interrupts more urgent than others.

Topic 3: PROGRAMMING LANGUAGES & TRANSLATORS (4 min)

- High-level: close to human language (Python, Java, C#). Portable across platforms. Need translation. Easier to write, read, debug.
- Low-level: close to machine code (assembly). Specific to processor architecture. Harder to write but more control and faster execution.
- Machine code: binary instructions executed directly by CPU. Not human-readable.

Three translators:

Feature	Compiler	Interpreter	Assembler	
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Translates	High-level → machine code	High-level → machine code	Assembly → machine
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How	Whole program at once	Line by line	One-to-one
Output	Object code/executable	No separate output	Machine code
Errors	All reported at end	Stops at first error	Direct translation
Speed	Slower to translate, fast run	Fast start, slower run	Fast
Needs source?	No (has .exe)	Yes (source needed each)	N/A

Topic 4: IDE FEATURES (2 min)

- Editor: write and edit code with syntax highlighting
- Error diagnostics: highlights syntax errors in real-time
- Run-time environment: execute and test code
- Auto-completion: suggests code completions
- Breakpoints: pause execution to inspect variables
- Step-through: execute line by line for debugging
- Translator integration: compile/interpret within IDE
- Prettyprint/reformat: auto-indents and formats code

12–25 min | GUIDED PRACTICE

Worked question: "Describe four functions of an operating system." (4 marks)

Students suggest; teacher writes best answer on board.

Second worked question: "Explain the difference between compilation and interpretation." (4 marks)

Pair-share → class response using comparison table above.

"A programmer is deciding whether to use a compiler or interpreter. Give one advantage of each for a beginner programmer." (2 marks)

25–35 min | INDEPENDENT PRACTICE

Students begin WORKSHEET 11A. Teacher circulates, identifies common misconceptions.

35–40 min | PLENARY + HOMEWORK

Plenary: 5-question true/false rapid-fire:

39. An OS manages memory. (T)
40. An interpreter translates the whole program before execution. (F)
41. Assembly language is a high-level language. (F)
42. Firmware is stored in ROM. (T)
43. A compiler produces an executable file. (T)

Homework: Complete WORKSHEET 11A. Research: find three different IDEs and list their features.

EAL Support:

- Bilingual keyword sheet with all software/internet terms
- Visual diagram showing compilation vs interpretation flowchart
- Sentence starters for OS function descriptions: "One function of an OS is..."

- Coloured highlighting for key terms in guided practice answers
- Pair EAL students with same-L1 peers where possible

Differentiation:

Emerging: Pre-completed table of OS functions; match function to description exercise; supported by TA; focus on recall questions (Q1-Q3)

Developing: Complete worksheet independently; describe 4 OS functions; compare compiler

vs interpreter using structured table

Secure: Complete all questions including evaluation (Q9); explain interrupts with priority levels; peer-teach; attempt 6-mark "explain why" questions

Resources Needed:

- Projector + slides
- Bilingual keyword sheets
- Comparison table template (blank + completed)
- WORKSHEET 11A
- True/false cards

--- LESSON 11.2 ---

LESSON 11.2 — The Internet, Digital Currency & Cyber Security

Topic: Internet structure, protocols, digital currency, malware, protections

Duration: 40 minutes

Lesson Type: Consolidation / Exam Practice

Syllabus Ref: 1.1, 3.2, 3.3, 3.4 (CIE 0478)

Learning Objective:

By the end of this lesson, students will be able to explain how the Internet and WWW differ, describe DNS and protocols, evaluate digital currencies, and describe all cyber security threats and protections.

Key Vocabulary:

Internet — 互联网 hù lián wǎng

World Wide Web — 万维网 wàn wéi wǎng

URL — 统一资源定位符 tǒng yī zī yuán dìng wèi fú

DNS — 域名系统 yù míng xì tǒng

HTTP — 超文本传输协议 chāo wén běn chuán shū xié yì

HTTPS — 安全超文本传输协议 ān quán chāo wén běn chuán shū xié yì

IP address — IP 地址 IP dì zhǐ

router — 路由器 lù yóu qì

blockchain — 区块链 qū kuài liàn

cryptocurrency — 加密货币 jiā mì huò bì

malware	— 恶意软件 è yì ruǎn jiàn
virus	— 病毒 bìng dú
worm	— 蠕虫 rú chóng
Trojan	— 木马 mù mǎ
spyware	— 间谍软件 jiān dié ruǎn jiàn
phishing	— 网络钓鱼 wǎng luò diào yú
firewall	— 防火墙 fáng huǒ qiáng
two-factor auth.	— 双因素认证 shuāng yīn sù rèn zhèng
biometric	— 生物识别的 shēng wù shí bié de

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"Write down everything you know about keeping data safe online. Minimum 3 things."

Collect responses: passwords, encryption, antivirus, don't click links, 2FA, etc.

Praise existing knowledge.

3–12 min | DIRECT INSTRUCTION

Topic 1: INTERNET vs WWW (2 min)

- Internet: global network of connected computers (infrastructure)
- WWW: system of interlinked hypertext documents accessed via browsers (service)
- Analogy: Internet = roads; WWW = traffic on roads; other services (email, FTP) also use roads
- URL structure: protocol://domain/path e.g., <https://www.example.com/page.html>

Topic 2: DNS, IP ADDRESSES & ROUTERS (3 min)

- IP address: unique identifier for each device on a network (IPv4: 4 bytes, e.g., 192.168.1.1; IPv6: 16 bytes for more addresses)
- DNS (Domain Name System): translates human-readable domain names to IP addresses
 - Hierarchical: root → TLD (.com, .cn) → domain → subdomain
 - DNS servers store these mappings worldwide
- Router: forwards data packets between networks; uses routing tables to decide best path; each network has a gateway router

Topic 3: HTTP vs HTTPS (1 min)

- HTTP: unsecured protocol for transferring web pages
- HTTPS: HTTP + SSL/TLS encryption; uses port 443; shows padlock icon; essential for sensitive data (banking, passwords)

Topic 4: DIGITAL CURRENCY (2 min)

- Digital currency: electronic money not physically held
- Blockchain: distributed digital ledger of transactions across many computers
 - Each block contains transaction data + hash of previous block

- Decentralised: no single authority controls it
- Transparent: all participants can see transactions
- Immutable: once recorded, cannot be changed
- Cryptocurrency: uses cryptography for security; Bitcoin, Ethereum
- e-CNY (China's digital yuan): central bank digital currency (CBDC); issued and controlled by People's Bank of China; NOT decentralised like Bitcoin
- Digital currency not using blockchain: some CBDCs, PayPal balance, store credit

Topic 5: CYBER SECURITY (4 min)

MALWARE TYPES:

- Virus: attaches to legitimate files; needs user action to spread; corrupts data
- Worm: standalone; self-replicates across networks; no user action needed
- Trojan: disguised as legitimate software; creates backdoor for attackers
- Spyware: secretly monitors user activity; keyloggers record keystrokes
- Adware: displays unwanted advertisements
- Ransomware: encrypts files; demands payment for decryption key

PROTECTIONS:

- Firewall: monitors/filter network traffic; blocks unauthorised access
- Antivirus: detects, quarantines, removes malware; uses signature + heuristic detection
- Anti-spyware: specifically targets spyware
- 2FA: requires two forms of identification (password + code/SMS/biometric)
- Biometric: fingerprint, facial recognition, iris scan
- Encryption: scrambles data so unauthorised parties cannot read it
- Access levels: different users have different permissions
- Physical security: locks, CCTV on server rooms
- Backup: regular copies of data stored separately
- User education: training users to recognise phishing, avoid suspicious downloads
- Penetration testing: authorised simulated attacks to find vulnerabilities

12–25 min | GUIDED PRACTICE

"A user receives an email claiming to be from their bank, asking them to click a link and enter their password."

Class discussion questions:

- What type of attack is this? (Phishing)
- How could the user protect themselves? (Check sender address, hover over link, don't click, contact bank directly, use 2FA)
- What technical protections could the bank implement? (SSL/TLS, fraud detection, email authentication, user education)

"Explain the role of DNS when you type www.google.com into your browser." (4 marks)

Students write; teacher reveals mark scheme.

25–35 min | INDEPENDENT PRACTICE

Students complete WORKSHEET 11B.

35–40 min | PLENARY + HOMEWORK

Plenary: "Malware Match" — show descriptions, students call out malware type:

- 44. Self-spreads without user action (Worm)
- 45. Needs you to open an attachment (Virus)
- 46. Looks like a game but steals passwords (Trojan)
- 47. Records everything you type (Keylogger/spyware)
- 48. Locks your files until you pay (Ransomware)

Homework: Complete WORKSHEET 11B. Create a poster: "5 Ways to Stay Safe Online" (suitable for Year 7 students).

EAL Support:

- Bilingual cyber security vocabulary cards
- Visual icons for each malware type
- Sentence frames: "This is an example of [malware] because..."
- Pre-written notes on digital currency for reference
- Video clip option: short animation on how DNS works (3 min)

Differentiation:

Emerging: Match malware to description; label diagram of DNS process; supported by TA; focus on Q1–Q4; provide word bank

Developing: Complete standard questions independently; explain DNS process in own words; describe two protection methods

Secure: Evaluate different types of digital currency; compare blockchain vs centralised banking; explain why 2FA is more secure than password alone; peer-teach malware types; complete all stretch questions

Resources Needed:

- Projector + slides with malware icons
- Bilingual keyword sheets
- DNS diagram (animated or static)
- WORKSHEET 11B
- Malware description cards

--- WORKSHEET 11A ---

WORKSHEET 11A: System Software, Languages & Translators

Student Name: _____ Date: _____

SECTION A: SHORT ANSWER

Q1. Give two differences between system software and application software. [2]

Q2. State four functions of an operating system. [4]

Q3. What is the difference between:

- a) High-level and low-level programming languages? [2]

b) Source code and object code? [2]

Q4. Complete the table comparing a compiler and an interpreter: [4]

Feature	Compiler	Interpreter
Translates		
Line-by-line?		
Produces .exe?		
Reports all errors?		

Q5. a) What is an interrupt? [1]

b) Give two sources of interrupts. [2]

SECTION B: EXAM-STYLE QUESTIONS

Q6. Explain why utility software is needed even when an operating system is installed. Give two examples of utility software and describe what each does. [4]

Q7. Describe three features of an Integrated Development Environment (IDE) and explain how each helps a programmer. [6]

Q8. A student wants to write a program that will run on both Windows and macOS without modification.

a) Should they use a compiler or interpreter? Justify your answer. [2]

b) Would a high-level or low-level language be more suitable? Explain why. [2]

Q9. Discuss the advantages and disadvantages of using a compiler versus an interpreter when developing a new program. [6]

--- WORKSHEET 11B ---

WORKSHEET 11B: The Internet, Digital Currency & Cyber Security

Student Name: _____ Date: _____

SECTION A: RECALL QUESTIONS

Q1. Explain the difference between the Internet and the World Wide Web. [2]

Q2. What is the purpose of DNS? [2]

Q3. State the difference between HTTP and HTTPS. [2]

Q4. State the purpose of a router. [1]

Q5. What is an IP address? [1]

SECTION B: DIGITAL CURRENCY

Q6. a) What is blockchain? [2]

b) Give two features of blockchain. [2]

Q7. Explain the difference between cryptocurrency (like Bitcoin) and a central bank digital currency (like e-CNY). [3]

Q8. Give two advantages and two disadvantages of using cryptocurrency for online payments. [4]

SECTION C: CYBER SECURITY

Q9. Complete the table describing each type of malware: [6]

Malware Type	How it spreads	Effect on system
Virus		
Worm		
Trojan		
Spyware		
Ransomware		

Q10. Describe four methods individuals or organisations can use to protect their data and systems from cyber attacks. [8]

Q11. A school stores student records on a central server.

- Explain why the school should use access levels. [2]
- Describe two other security measures the school should implement. [4]

Q12. Explain the process that occurs when a user types a URL into their web browser. Your answer should include the roles of DNS, IP addresses, routers, and servers. [6]

--- ANSWER KEY (WEEK 11) ---

WORKSHEET 11A — ANSWER KEY

Q1. Any two from:

- System software manages the computer hardware; application software performs tasks for the user (1)
- System software runs in background; application software is user-facing (1)
- Examples differ: OS/utility/drivers vs Word/games/browser (1)

(2 marks max)

Q2. Any four from (1 each, max 4):

- Memory management
- File management
- I/O management (device driver management)
- Interrupt handling
- Security (user accounts/passwords)
- Providing user interface (GUI/CLI)
- Processor/multitasking scheduling
- Error reporting
- Loading and executing programs

Q3. a) High-level languages are closer to human language / English-like (1) and are portable across different platforms (1).

Low-level languages are closer to machine code / specific to a processor (1).

High-level needs translation; low-level gives more control over hardware (1).

(Max 2)

b) Source code is the original code written by the programmer in a high-level language (1). Object code is the machine code output produced by a compiler (1).

Q4. (1 mark per correct cell, max 4)

Feature	Compiler	Interpreter	
Translates	High-level → machine code	High-level → machine	
Line-by-line?	No (whole program)	Yes	
Produces .exe?	Yes	No	
Reports all errors?	Yes (at end)	No (stops at first)	

Q5. a) An interrupt is a signal sent to the CPU to request immediate attention (1).

b) Any two from: hardware error/failure, user input (keyboard/mouse), timer signal, I/O device request, power failure (1 each, max 2).

Q6. Utility software performs specific maintenance/security tasks that the OS does not provide (1). Examples (1 each, max 2): antivirus (detects/removes malware), disk defragmenter (reorganises files for faster access), backup software, file compression, disk formatter. Description of each utility (1 each, max 2).

Q7. Any three from (feature 1 + explanation 1, max 6):

- Editor with syntax highlighting: helps programmers spot mistakes and write code more easily (1+1)
- Error diagnostics: highlights syntax errors as code is typed, saving time (1+1)
- Auto-completion: suggests code completions, reducing typing errors and speeding up coding (1+1)
- Breakpoints: allows programmer to pause execution at specific lines to check variable values (1+1)
- Step-through: executes code line by line to trace logic and find bugs (1+1)
- Translator integration: compile/run code without leaving the IDE (1+1)
- Prettyprint: auto-formats code for readability (1+1)
- Run-time environment: test programs directly within the IDE (1+1)

Q8. a) Either can be acceptable with good justification.

Compiler: produces standalone executable that can be distributed to both platforms (if cross-compiler used) (1); runs without source code on target machine (1).

OR Interpreter: interpreted languages (Python) are often portable across platforms without recompilation (1); source code can run on any system with the interpreter installed (1).

b) High-level language (1): it is portable across platforms and easier to write and debug (1).

Q9. Mark scheme (6 marks):

Compiler advantages (1 each, max 2):

- Produces standalone executable — source code not needed to run
- Program runs faster after compilation
- All errors reported at once at end
- Code optimised during compilation

Interpreter advantages (1 each, max 2):

- Easier to debug — stops at first error
- Faster to test — no compile step needed
- Source code needed each time (portable across platforms with interpreter)

Disadvantages of each implied by opposite advantages (1 each, max 2)

Overall balanced discussion with conclusion = 1 extra

(Max 6 — award best 6 points)

WORKSHEET 11B — ANSWER KEY

Q1. The Internet is the global network infrastructure of connected computers (1).

The World Wide Web is a service/system of interlinked hypertext documents that runs on top of the Internet, accessed through browsers (1).

Q2. DNS (Domain Name System) translates human-readable domain names (e.g., google.com)

into IP addresses (1) that computers use to locate servers on the Internet (1).

Q3. HTTP is unsecured data transmission (1); HTTPS uses SSL/TLS to encrypt data between browser and server, providing security (1).

Q4. A router forwards data packets between networks (1) and determines the best path for data using routing tables (accept: connects networks together).

Q5. An IP address is a unique numerical identifier assigned to each device on a network (1).

Q6. a) Blockchain is a distributed digital ledger that records transactions across many computers (1) in a way that makes the record difficult to alter (1).

b) Any two from: decentralised (no single authority), transparent (all can see), immutable (cannot be changed once recorded), uses cryptography for security, consensus mechanism for validation (1 each, max 2).

Q7. • Cryptocurrency (e.g., Bitcoin) is decentralised — no central authority controls it (1)

- CBDC (e.g., e-CNY) is centralised and issued by the country's central bank (1)
- Cryptocurrency value fluctuates; CBDC is typically pegged to national currency (1)
- Cryptocurrency is typically anonymous; CBDC transactions may be traceable by government (1)

(Max 3 — award best 3 points)

Q8. Advantages (1 each, max 2): fast/cross-border transactions, lower fees than banks, no bank account needed, privacy/pseudonymity, decentralised.

Disadvantages (1 each, max 2): price volatility, not widely accepted, security risks (hacking/wallets lost), lack of consumer protection, environmental concerns (mining).

Q9. (1 mark per correct cell, max 6 — allow similar correct descriptions)

Malware Type	How it spreads	Effect on system
Virus	Attaches to files; needs user action to spread	Corrupts/deletes files
Worm	Self-replicates across networks automatically	Slows network, uses bandwidth
Trojan	Disguised as legitimate software	Creates backdoor, steals data
Spyware	Bundled with other software or downloads	Monitors activity, steals personal info
Ransomware	Phishing emails, malicious downloads	Encrypts files, demands payment for decryption

Q10. Any four from (method 1 + description 1, max 8):

- Firewall: monitors and filters incoming/outgoing network traffic to block unauthorised access (1+1)
- Antivirus: detects, quarantines, and removes malware (1+1)
- 2FA: requires two forms of identification (password + code/biometric) (1+1)
- Encryption: scrambles data so it cannot be read if intercepted (1+1)
- Access levels: different users have different permissions (1+1)
- Physical security: locks, CCTV on server rooms (1+1)
- Regular backups: copies of data stored separately for recovery (1+1)
- Penetration testing: simulated attacks to find vulnerabilities (1+1)
- User education: training users to recognise phishing and avoid threats (1+1)
- Biometric authentication: fingerprint/facial recognition for strong identity verification (1+1)

Q11. a) Access levels ensure only authorised staff can access sensitive student data (1), reducing risk of data breaches / protecting privacy (1).

b) Any two from (method 1 + description 1, max 4):

- Firewall to prevent unauthorised network access (1+1)
- Encryption of stored data so it cannot be read if stolen (1+1)
- Regular backups to recover from ransomware/accidental deletion (1+1)
- Antivirus software to detect malware (1+1)
- Strong password policy / 2FA for all accounts (1+1)
- Physical security (locked server room, CCTV) (1+1)

Q12. Mark scheme (6 marks):

- User types URL (domain name) into browser (1)
- Browser sends request to DNS server (1)

- DNS looks up domain name and returns the corresponding IP address (1)
- Browser uses the IP address to send a request packet (1)
- Router(s) forward the packet across networks towards the destination server (1)
- Web server receives the request and sends the web page back (1)
- Data travels back via routers to the user's browser (1)

(Max 6 — award best 6 points)

WEEK 12: PAPER 1 REVIEW — EMERGING TECH & MIXED PRACTICE

LESSON 12.1 — Emerging Technologies, Network Hardware & Ethics

Topic: Automated systems, AI, network hardware, environmental/ethical issues

Duration: 40 minutes

Lesson Type: Consolidation / Exam Practice

Syllabus Ref: 1.5, 2.2, 3.1, 3.5 (CIE 0478)

Learning Objective:

By the end of this lesson, students will be able to explain automated systems and AI applications, describe network hardware components, and discuss environmental and ethical considerations of computing.

Key Vocabulary:

automated system — 自动化系统 zì dòng huà xì tǒng

monitoring — 监测 jiān cè

control — 控制 kòng zhì

artificial intelligence — 人工智能 réngōng zhì néng

machine learning — 机器学习 jī qì xué xí

robotics — 机器人技术 jī qì rén jì shù

NIC — 网卡 wǎng kǎ

MAC address — MAC 地址 MAC dì zhǐ

topology — 拓扑结构 tuò pǔ jié gòu

star topology — 星型拓扑 xīng xíng tuò pǔ

bus topology — 总线拓扑 zǒng xiàn tuò pǔ

ethics — 伦理 lún lǐ

e-waste — 电子垃圾 diàn zǐ lā jī

carbon footprint — 碳足迹 tàn zú jì

digital divide — 数字鸿沟 shù zì hóng gōu

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"List three ways AI is used in everyday life."

Collect: voice assistants, recommendations, navigation, facial recognition, etc.

Also ask: "Name one positive and one negative impact of AI on jobs."

3–12 min | DIRECT INSTRUCTION

Topic 1: AUTOMATED SYSTEMS (2 min)

- Monitoring system: measures data, displays it, takes NO action

Example: weather station displays temperature

- Control system: measures data AND takes action based on it

Example: thermostat measures temperature → turns heater on/off

- Microprocessor (embedded system) reads sensor → processes data → sends signal to actuator. Feedback loop: output affects next input reading.

- Advantages: works 24/7, consistent, faster response, dangerous environments, reduces human error

- Disadvantages: expensive setup, maintenance needed, no human judgment, job displacement, system failures can be serious

Topic 2: AI & MACHINE LEARNING (3 min)

- AI: computer systems that perform tasks requiring human intelligence (reasoning, learning, problem-solving, understanding language, perception)

- Machine learning: subset of AI where systems learn from data without being explicitly programmed

- Training data fed to algorithm → model recognises patterns → makes predictions
- Supervised: labelled training data; unsupervised: finds patterns in unlabelled data

- Applications: voice recognition (Siri/Alexa), image recognition, medical diagnosis, fraud detection, recommendation systems, autonomous vehicles, game playing

- Ethics: bias in training data, privacy concerns, job displacement, decision transparency ("black box" problem), accountability for AI decisions

Topic 3: NETWORK HARDWARE (2 min)

- NIC (Network Interface Card): connects device to network; every NIC has unique MAC address (48-bit, hexadecimal, e.g., 00:1A:2B:3C:4D:5E); used on local network
- MAC vs IP: MAC = physical/hardware address (fixed); IP = logical address (can change)
- Topologies:

Star: all devices connect to central switch/hub. Easy to add devices; one cable failure doesn't affect others; switch failure = whole network down; more cabling.

Bus: all devices connect to single cable (backbone). Simple, cheap; cable failure = whole network down; difficult to add devices; collisions slow network.

Mesh: devices connected to multiple others. Very reliable; expensive; complex.

- Switch: intelligent — sends data only to intended recipient (uses MAC addresses)
- Hub: broadcast — sends data to all connected devices (less efficient)

Topic 4: ENVIRONMENTAL & ETHICAL CONSIDERATIONS (2 min)

Environmental:

- E-waste: discarded electronics contain toxic materials; recycling needed
- Energy consumption: data centres use large amounts of electricity
- Carbon footprint: manufacturing and running IT equipment produces CO₂
- Reduce: cloud computing can be more efficient than individual servers
- Recycling schemes, WEEE directive (Europe), longer product lifespans

Ethical:

- Digital divide: unequal access to technology between countries/regions/socioeconomic

groups; affects education and employment opportunities

- Privacy: data collection by companies and governments; surveillance; cookies
- Censorship: controlling internet access and content
- AI ethics: bias, job losses, autonomous weapons, deepfakes
- Software licensing: piracy, open-source vs proprietary, GPL

12–25 min | GUIDED PRACTICE

"A factory uses robots to assemble cars. Explain two advantages and two disadvantages of this automated system." (4 marks)

Students discuss in pairs; teacher selects best answers and adds missing points.

"Describe the difference between a monitoring system and a control system. Give an example of each." (4 marks)

"Explain why AI systems can show bias." (2 marks)

— Because training data may not be representative / may contain human biases.

25–35 min | INDEPENDENT PRACTICE

Students complete WORKSHEET 12A.

35–40 min | PLENARY + HOMEWORK

Plenary: "Ethics Debate" — pose statement: "AI should never be used to make decisions about people's jobs." Students vote agree/disagree, then 2 students justify each side.

Homework: Complete WORKSHEET 12A. Research: find one recent news story about AI ethics and write a 5-line summary.

EAL Support:

- Bilingual glossary with all technology terms
- Visual diagram of monitoring vs control system with arrows
- Sentence starters for ethics questions: "One ethical issue is..."
- Pre-written key points for AI applications (reference card)

- Network topology diagrams with labels in both languages

Differentiation:

Emerging: Labelled diagrams provided; match network device to function; supported by TA; sentence starters for all written answers; focus on Q1–Q4

Developing: Complete standard worksheet independently; explain monitoring vs control with examples; describe two ethical issues

Secure: Evaluate environmental impact of computing trends; debate AI ethics with balanced argument; compare all three network topologies with detailed pros/cons; complete all stretch questions; peer-teach

Resources Needed:

- Projector + slides
- Bilingual keyword sheets
- Network topology diagrams
- Monitoring/control system diagrams
- WORKSHEET 12A
- Debate statement slide

--- LESSON 12.2 ---

LESSON 12.2 — Mixed Paper 1 Practice (Timed)

Topic: All Paper 1 topics in timed exam conditions

Duration: 40 minutes

Lesson Type: Exam Practice / Timed Assessment

Syllabus Ref: All Paper 1 topics (CIE 0478)

Learning Objective:

By the end of this lesson, students will have practised Paper 1 questions under timed conditions, reviewed common mistakes, and learned exam technique for different question types.

Key Vocabulary:

exam technique — 考试技巧 kǎo shì jì qiǎo

time management — 时间管理 shí jiān guǎn lǐ

key term — 关键词 guān jiàn cí

mark allocation — 分数分配 fēn shù fēn pèi

common mistake — 常见错误 cháng jiàn cuò wù

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"In a 4-mark 'Explain' question, how many distinct points should you aim to make?"

Answer: at least 4 distinct points, each explained. One point = one mark.

3–5 min | EXAM TECHNIQUE BRIEFING

Quick tips for Paper 1:

- 1-mark questions: one fact, one sentence. Don't waste time.
- 2-mark "state/describe": two distinct points.
- 4-mark "explain": state + explain for each point (4 distinct points).
- 6-mark "discuss/explain in detail": 6 points with development.

Use PEEL: Point, Evidence, Explain, Link.

- ALWAYS read the command word: state, describe, explain, compare, evaluate, discuss, justify.
- If stuck, move on — come back later. Don't leave blanks (guess if needed).
- For comparison questions: use "whereas" to show contrast.

5–35 min | TIMED PRACTICE

Students complete WORKSHEET 12B — timed 30-minute section.

Teacher circulates; reminds students of time remaining (15 min, 5 min warnings).

Students work in exam conditions: no talking, no phones, no notes.

35–40 min | PLENARY + COMMON MISTAKES

Quick review of answers to 12B — go through mark scheme rapidly.

Common mistakes to highlight:

- 49. Confusing RAM and ROM properties (volatile vs non-volatile)
- 50. Forgetting that USB is serial, not parallel
- 51. Saying "the CPU fetches data" without naming the register
- 52. Describing what a firewall does but not how it works
- 53. Missing the difference between Internet and WWW
- 54. Not explaining "why" in explain questions — just stating facts

Homework: Review all Paper 1 topics using the CIE specification. Use the gap analysis table to identify weak areas.

EAL Support:

- Exam technique card with command words explained in Chinese
- Extra time consideration for EAL students (discuss with SENCO)
- Clarify command words before timed section begins
- Allow bilingual dictionary access during practice (not in real exam)

Differentiation:

Emerging: Shorter timed section (15 min); fewer questions; word bank provided; can use notes for reference (scaffolded exam practice)

Developing: Complete standard 30-min timed section; attempt all questions

Secure: Complete 30-min section plus extension questions; act as "exam helper" by reviewing peers' answers afterwards

Resources Needed:

- WORKSHEET 12B (printed, enough for all students)
- Timer (visible to all)

- Mark scheme printed for teacher reference
- "Exam command words" poster

--- WORKSHEET 12A ---

WORKSHEET 12A: Emerging Technologies, Network Hardware & Ethics

Student Name: _____ Date: _____

SECTION A: AUTOMATED SYSTEMS & AI

Q1. Describe the difference between a monitoring system and a control system.

Give an example of each. [4]

Q2. A farmer uses an automated irrigation system in a greenhouse.

- Name one sensor and one actuator that would be used. [2]
- Explain how the system works as a control system. [3]

Q3. Explain what is meant by machine learning. [2]

Q4. Give two applications of artificial intelligence and explain how each benefits users. [4]

Q5. Explain why an AI system might produce biased results. [2]

SECTION B: NETWORK HARDWARE

Q6. a) What is a MAC address? [1]

- How does a MAC address differ from an IP address? [2]

Q7. Describe the star topology. Include one advantage and one disadvantage. [3]

Q8. Explain the difference between a switch and a hub. [2]

SECTION C: ENVIRONMENTAL & ETHICAL

Q9. Describe two ways that the production and use of computers can damage the environment. [4]

Q10. Explain what is meant by the "digital divide" and describe one way it can be reduced. [3]

Q11. A company is developing an AI system to screen job applications.

Discuss the ethical issues this raises. [6]

--- WORKSHEET 12B ---

WORKSHEET 12B: Mixed Paper 1 Practice — TIMED SECTION (30 minutes)

Student Name: _____ Date: _____

Instructions: You have 30 minutes. Answer ALL questions. Write in full sentences.

Q1. The Internet and the World Wide Web are often confused.

Explain the difference between them. [2]

Q2. A student is sending a large video file to a friend using the Internet.

- Explain how the file is transmitted using packet switching. [4]
- Give two advantages of packet switching. [2]

- Q3. Describe the Fetch-Decode-Execute cycle. Your answer should refer to at least three registers. [6]
- Q4. A computer has the following memory specifications: 8 GB RAM, 256 GB SSD, 4 MB cache.
- a) State which of these is volatile memory. [1]
 - b) Explain the purpose of cache memory. [2]
 - c) Explain why an SSD is more suitable than an HDD for a laptop that will be carried around. [2]
- Q5. Explain the role of the operating system in managing memory. [3]
- Q6. A company network uses a firewall.
- a) Describe how a firewall protects the network. [2]
 - b) State two other security measures the company should use. [2]
- Q7. Compare the use of a compiler and an interpreter. [4]
- Q8. Describe two ethical issues related to the increasing use of artificial intelligence in the workplace. [4]
- Q9. A school is replacing its old magnetic hard drives with solid-state storage. Evaluate this decision. Consider both advantages and disadvantages. [6]

--- ANSWER KEY (WEEK 12) ---

WORKSHEET 12A — ANSWER KEY

- Q1. A monitoring system measures data and displays it but takes no action (1).
Example: weather station displays temperature (1).
A control system measures data AND takes action based on the readings (1).
Example: thermostat controls heating (1).
- Q2. a) Sensor: temperature sensor / humidity sensor / soil moisture sensor (1)
Actuator: water pump / valve / sprinkler (1)
- b) • Sensor measures soil moisture/temperature (1)
- Microprocessor compares reading to preset value (1)
 - If below threshold, signal sent to actuator (pump) to release water (1)
- (Accept: feedback loop description)
- Q3. Machine learning is a type of AI where computer systems learn from data (1) without being explicitly programmed / by identifying patterns in data (1).
- Q4. Any two from (application 1 + benefit 1, max 4):
- Voice assistants (Siri/Alexa): help users control devices hands-free, accessibility for disabled users (1+1)
 - Medical diagnosis: analyse scans/images to detect diseases earlier and more accurately than humans (1+1)

- Fraud detection: banks use AI to identify unusual transactions, protecting customers (1+1)
- Recommendation systems: suggest products/content users will like, improving user experience (1+1)
- Autonomous vehicles: reduce human error in driving, potentially fewer accidents (1+1)
- Image recognition: security/social media tagging (1+1)

Q5. AI systems learn from training data (1). If the training data contains biases or is not representative of all groups, the AI will learn and reproduce those biases in its decisions (1).

Q6. a) A MAC address is a unique hardware identifier assigned to a NIC (1).

b) MAC address is physical/fixed (burned into NIC) and used on local network (1);
IP address is logical/can change and used across networks/Internet (1).

Q7. In a star topology, all devices are connected to a central switch/hub (1).

Advantage: easy to add new devices / one cable failure doesn't affect others (1).

Disadvantage: if the central switch fails, the whole network goes down / uses more cabling than bus (1).

Q8. A switch sends data only to the intended recipient (using MAC addresses) (1);
a hub broadcasts data to all connected devices (1).

Q9. Any two from (issue 1 + explanation 1, max 4):

- E-waste: discarded computers contain toxic materials (lead, mercury) that pollute soil and water if not properly recycled (1+1)
- Energy consumption: data centres and devices use large amounts of electricity, often from fossil fuels, contributing to climate change (1+1)
- Manufacturing: production of chips and devices uses scarce resources and produces pollution (1+1)
- Carbon footprint: transportation of devices globally produces CO2 emissions (1+1)

Q10. The digital divide is the inequality in access to technology and the Internet between different groups or regions (1). It can be reduced by: government investment in infrastructure, providing subsidised devices to low-income families, public access centres (libraries), digital literacy training (1 for method + 1 for development).

Q11. Mark scheme (6 marks):

- Bias: AI may discriminate against certain groups if training data is biased (1+1)
- Job displacement: AI screening may replace human HR staff, causing unemployment (1+1)
- Privacy: AI analyses personal data, raising privacy concerns (1+1)
- Transparency: AI decisions may not be explainable ("black box"), making it hard

to challenge unfair decisions (1+1)

- Accountability: unclear who is responsible if AI makes a discriminatory decision (1+1)
- Consent: applicants may not know AI is screening their application (1+1)

(Max 6 — award best 6 points with at least 3 issues discussed)

WORKSHEET 12B — ANSWER KEY

Q1. The Internet is the global network infrastructure of connected computers (1).

The World Wide Web is a service of interlinked documents/pages accessed via browsers, running on the Internet (1).

Q2. a) Mark scheme (4 marks):

- File is split into packets (1)
- Each packet has a header with source/destination address and sequence number (1)
- Packets travel independently across the network via routers (1)
- Packets are reassembled at the destination in the correct order (1)

b) Any two from: no dedicated line needed, fault tolerant (one route fails others work), efficient use of bandwidth, packets can take optimal routes (1 each, max 2).

Q3. Mark scheme (6 marks):

- PC holds address of next instruction (1)
- Address copied to MAR (1)
- Address bus carries address to memory; control bus sends READ (1)
- Instruction/data placed on data bus and copied to MDR (1)
- Instruction copied from MDR to CIR (1)
- PC increments (1)
- Instruction decoded by control unit (1)
- Appropriate execution (ALU for calculations, etc.) (1)
- Result stored in ACC (1)

(Max 6 — must name at least 3 registers for full marks)

Q4. a) RAM (1)

b) Cache stores frequently accessed data/instructions (1) so CPU doesn't need to access slower RAM as often, speeding up processing (1).

c) SSD has no moving parts (1) so is more durable/resistant to physical shock when carried around / faster / lighter / uses less power (1).

Q5. OS allocates RAM to programs (1), manages virtual memory (using hard disk as extra RAM) (1), and keeps track of which memory locations are free and in use (1).

Q6. a) A firewall monitors incoming and outgoing network traffic (1) and blocks unauthorised access based on rules (1).

b) Any two from: antivirus, 2FA, encryption, access levels, physical security, regular backups, penetration testing, user education (1 each, max 2).

Q7. Any four from (1 each, max 4):

- Compiler translates entire program at once; interpreter line by line
- Compiler produces standalone executable; interpreter needs source code each time
- Compiler reports all errors at end; interpreter stops at first error
- Compiled program runs faster; interpreted program starts faster
- Compiler produces object code file; interpreter does not
- Compiled code can run without source; interpreted needs source or bytecode

Q8. Any two from (issue 1 + explanation 1, max 4):

- Job displacement: AI automation may replace human workers, causing unemployment

in certain sectors (1+1)

- Bias and discrimination: AI systems may make biased decisions affecting hiring, lending, or law enforcement (1+1)
- Privacy: AI surveillance and data analysis infringe on personal privacy (1+1)
- Lack of accountability: unclear who is responsible when AI makes harmful decisions (1+1)
- Loss of human skills: over-reliance on AI may reduce human expertise (1+1)

Q9. Mark scheme (6 marks):

Advantages of SSD (1 each, max 3):

- Faster read/write speeds — programs load quicker
- No moving parts — more durable, less likely to fail from shock
- Uses less power — longer battery life for laptops
- Silent operation — no noise from spinning disks
- Lighter weight — better for portable devices

Disadvantages of SSD (1 each, max 3):

- More expensive per GB than HDD — higher cost for school
- Lower maximum capacity than HDD (at same price point)
- Limited write cycles — may wear out over time (though modern SSDs manage this)
- Data recovery harder if drive fails

Balanced conclusion with justified recommendation = 1 extra mark

(Max 6 — need both advantages and disadvantages for top marks)

WEEK 13: PAPER 2 REVIEW — ALGORITHMS & PROGRAMMING

LESSON 13.1 — Algorithms: Pseudocode, Flowcharts, Searching & Sorting

Topic: Algorithm design, standard symbols, searching, bubble sort, trace tables

Duration: 40 minutes

Lesson Type: Consolidation / Exam Practice

Syllabus Ref: 2.1, 2.2 (CIE 0478)

Learning Objective:

By the end of this lesson, students will be able to write and trace pseudocode, use standard flowchart symbols, explain when to use different search algorithms, and trace bubble sort with a trace table.

Key Vocabulary:

algorithm	— 算法 suàn fǎ
pseudocode	— 伪代码 wěi dài mǎ
flowchart	— 流程图 liú chéng tú
decomposition	— 分解 fēn jiě
abstraction	— 抽象 chōu xiàng
linear search	— 线性搜索 xiàn xìng sōu suǒ
binary search	— 二分搜索 èr fēn sōu suǒ
bubble sort	— 冒泡排序 mào pào pái xù
trace table	— 追踪表 zhuī zōng biǎo
iteration	— 迭代 dié dài
selection	— 选择 xuǎn zé
efficiency	— 效率 xiào lǜ

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"Draw the flowchart symbol for: start/stop, process, decision, input/output. Label each one." (4 symbols). Students draw in books. Check answers rapidly.

3–12 min | DIRECT INSTRUCTION

Topic 1: PSEUDOCODE CONVENTIONS (2 min)

Cambridge pseudocode uses:

- INPUT / OUTPUT
- DECLARE variable_name : type (INTEGER, REAL, STRING, BOOLEAN, CHAR, DATE)
- CONSTANT name = value
- $\leftarrow$ for assignment: counter $\leftarrow$ counter + 1
- IF ... THEN ... ELSE ... ENDIF
- CASE OF ... OTHERWISE ... ENDCASE
- FOR ... TO ... NEXT
- WHILE ... DO ... ENDWHILE
- REPEAT ... UNTIL ...
- PROCEDURE name(parameters) / ENDPROCEDURE
- FUNCTION name(parameters) RETURNS type / ENDFUNCTION
- CALL name(parameters)
- // for comments
- MOD (modulo), DIV (integer division), LENGTH(), SUBSTRING(), ROUND()

- ARRAY declarations: DECLARE names : ARRAY[1:10] OF STRING

Topic 2: FLOWCHART SYMBOLS (1 min)

- Oval/rounded rectangle: Start / Stop (terminator)
- Rectangle: Process (calculation, assignment)
- Diamond: Decision (yes/no, true/false — always two exits)
- Parallelogram: Input / Output
- Arrows: Flow direction

Key rule: flowchart must have ONE start and ONE stop.

Topic 3: PROBLEM SOLVING (1 min)

- Decomposition: breaking a complex problem into smaller sub-problems

Example: "Make a game" → graphics, physics, scoring, menus, sound

- Abstraction: removing unnecessary detail to focus on what's important

Example: Maps show roads not buildings; user login abstracts away encryption details

- Why: makes problems manageable, allows team work, enables code reuse

Topic 4: SEARCHING ALGORITHMS (3 min)

LINEAR SEARCH:

- Checks each item in order from start to finish
- Works on sorted OR unsorted data
- Best case: 1 comparison (item at start)
- Worst case: n comparisons (item at end or not present)
- Time complexity: $O(n)$

BINARY SEARCH:

- Repeatedly divides sorted data in half
- ONLY works on SORTED data
- Compare target to middle item; if not found, search left or right half
- Best case: 1 comparison (item is middle)
- Worst case: $\log_2(n)$ comparisons
- Time complexity: $O(\log n)$ — much faster for large datasets

WHEN TO USE WHICH:

- Small dataset or unsorted data → linear search (simpler, no sorting needed)
- Large sorted dataset → binary search (much faster)

Topic 5: BUBBLE SORT (3 min)

- Compare adjacent pairs; swap if in wrong order
- After each pass, the largest unsorted item "bubbles up" to correct position
- Repeat for $n-1$ passes (where n = number of items)
- Can be optimised: stop if a pass makes no swaps (already sorted)
- Time complexity: $O(n^2)$ — slow for large datasets

Pseudocode:

FOR $i \leftarrow 1$ TO $n-1$

FOR $j \leftarrow 1$ TO $n-i$

```

IF list[j] > list[j+1] THEN
  temp ← list[j]
  list[j] ← list[j+1]
  list[j+1] ← temp
ENDIF
NEXT j
NEXT i

```

12–25 min | GUIDED PRACTICE

Activity 1: Trace bubble sort on [5, 3, 8, 1] — whole class on board.

Pass 1: 5>3 swap → [3,5,8,1]; 5<8 no swap; 8>1 swap → [3,5,1,8] ✓

Pass 2: 3<5 no swap; 5>1 swap → [3,1,5,8]; 5<8 no swap

Pass 3: 3>1 swap → [1,3,5,8]; 3<5 no swap — SORTED!

Complete trace table together:

Pass	List State	Swaps?
1	3,5,1,8	Yes
2	3,1,5,8	Yes
3	1,3,5,8	No

Activity 2: "When would you use linear search instead of binary search?"

Students discuss in pairs → class check.

25–35 min | INDEPENDENT PRACTICE

Students complete WORKSHEET 13A. Teacher circulates, checks trace tables for accuracy.

35–40 min | PLENARY + HOMEWORK

Plenary: "3-2-1" — 3 things remembered about algorithms, 2 differences between linear and binary search, 1 question still unclear.

Homework: Complete WORKSHEET 13A. Write pseudocode for linear search.

EAL Support:

- Bilingual keyword sheet with pseudocode terms
- Visual flowchart symbol reference card
- Pre-structured trace table templates
- Colour-coded pseudocode (syntax highlighting style)
- Sentence frames: "Linear search is used when..."

Differentiation:

Emerging: Pre-completed trace table with some values filled; match flowchart symbols to names; supported by TA; complete Q1–Q3 only

Developing: Complete trace tables independently; explain when to use each search; write simple pseudocode

Secure: Optimise bubble sort pseudocode; compare efficiency using Big-O notation; peer-teach; complete all stretch questions; explain decomposition and abstraction with complex examples

Resources Needed:

- Projector
- Flowchart symbol reference card
- Cambridge pseudocode reference sheet
- WORKSHEET 13A
- Trace table templates (optional)

--- LESSON 13.2 ---

LESSON 13.2 — Python Programming Review

Topic: Python syntax, data types, I/O, selection, iteration, nested structures, errors

Duration: 40 minutes

Lesson Type: Consolidation / Coding Practice

Syllabus Ref: 2.2 (CIE 0478)

Learning Objective:

By the end of this lesson, students will be able to write Python code using variables, all data types, input/output, selection, iteration, and nested structures; and will be able to identify and fix common errors.

Key Vocabulary:

variable	— 变量 biàn liàng
data type	— 数据类型 shù jù lèi xíng
string	— 字符串 zì fú chuàn
integer	— 整数 zhěng shù
float / real	— 浮点数 fú diǎn shù
Boolean	— 布尔值 bù ěr zhí
casting	— 类型转换 lèi xíng zhuǎn huàn
selection	— 选择结构 xuǎn zé jié gòu
iteration	— 循环结构 xún huán jié gòu
nested	— 嵌套的 qiàn tào de
syntax error	— 语法错误 yǔ fǎ cuò wù
logic error	— 逻辑错误 luó jí cuò wù
runtime error	— 运行时错误 yùn xíng shí cuò wù

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"This code has an error. Find and fix it:

```
age = input('Enter your age: ')
```

```
if age > 18:
```

```
    print('Adult')
```

Answer: input returns string — need int() cast: age = int(input('Enter your age: '))

3–12 min | DIRECT INSTRUCTION

Topic 1: VARIABLES & DATA TYPES (2 min)

- Variable: named storage location. Python is loosely typed (no DECLARE needed).
- Types: int, float, str, bool
- Casting: int("5") → 5; str(5) → "5"; float("3.14") → 3.14
- Naming: meaningful, no spaces, no keywords, case-sensitive
- Constants: convention is UPPER_CASE (Python doesn't enforce)

Topic 2: INPUT & OUTPUT (1 min)

- input("Prompt") — always returns STRING
- print("Hello", name) — comma adds space
- print(f"Hello {name}, you are {age}") — f-string (modern)
- print("Hello " + name) — concatenation (strings only)

Topic 3: OPERATORS (1 min)

- Arithmetic: +, -, *, /, // (integer div), % (modulo), ** (power)
- Comparison: ==, !=, <, >, <=, >=
- Logical: AND, OR, NOT (Python: and, or, not)
- String: + (concatenation), * (repetition)

Topic 4: SELECTION (2 min)

if condition:

```
    # code
```

elif condition:

```
    # code
```

else:

```
    # code
```

- Indentation is CRITICAL in Python (4 spaces)
- elif = "else if" — checks another condition
- Can be nested: if inside if

Topic 5: ITERATION (2 min)

FOR loop — definite iteration (know how many times):

```
for i in range(10):    # 0 to 9
```

```
for i in range(1, 11): # 1 to 10
```

```
for i in range(0, 10, 2): # 0, 2, 4, 6, 8
```

```
for name in names:    # iterate through list
```

WHILE loop — indefinite iteration (condition-controlled):

```
while count < 10:
```

```
# code
```

```
count = count + 1 # MUST change condition variable!
```

NESTED LOOPS: loop inside a loop. Common for 2D data.

```
for row in range(3):
```

```
    for col in range(3):
```

```
        print(row, col)
```

Topic 6: COMMON ERRORS (2 min)

- Syntax error: misspelling, missing colon, wrong indentation

→ Python shows line number and error type

- Logic error: code runs but wrong output

→ Use print() statements to trace variable values

- Runtime error: occurs during execution (e.g., dividing by zero, index out of range)

→ Use try-except or validate inputs

- Type error: mixing types (e.g., "5" + 3)

→ Cast to correct type

- Name error: variable used before assignment or misspelled
- Indentation error: wrong/mixed spaces and tabs

12-25 min | GUIDED PRACTICE

"Write Python code that asks for 5 numbers and prints the largest."

Build together:

```
largest = -999999 # or use first input
```

```
for i in range(5):
```

```
    num = int(input("Enter a number: "))
```

```
    if num > largest:
```

```
        largest = num
```

```
print("The largest number is:", largest)
```

"This code should print even numbers from 2 to 20. Fix it:"

```
for i in range(21):
```

```
    if i % 2 == 0:
```

```
        print(i)
```

Fixes: range(2, 21, 2) is better; == not =; 0 not needed.

Better version: for i in range(2, 21, 2): print(i)

25-35 min | INDEPENDENT PRACTICE

Students complete WORKSHEET 13B on computers or paper.

Teacher circulates, helps with debugging.

35-40 min | PLENARY + HOMEWORK

Plenary: "Bug Hunt" — show 3 code snippets with errors; students identify error type.

55. `print("Hello)` — Syntax (missing quote)

56. `x = 5 / 0` — Runtime (divide by zero)

57. `for i in range(3): print(i+1)` → prints 1,2,3 but wanted 0,1,2 — Logic

Homework: Complete WORKSHEET 13B. Fix 5 common Python errors (provided sheet).

EAL Support:

- Bilingual Python keyword reference card
- Colour-coded code examples (keywords, strings, variables in different colours)
- Common error messages translated to Chinese
- Sentence frames for explaining code: "This line... because..."
- Visual indentation guides

Differentiation:

Emerging: Code skeletons with blanks to fill; supported by TA; focus on Q1–Q4

(basic syntax); error-spotting with hints

Developing: Write complete programs independently; debug code with minimal support;

complete Q1–Q7; explain what their code does

Secure: Write efficient solutions; complete all questions including stretch;

peer-teach; optimise given code; explain differences between for and while loops with specific use cases

Resources Needed:

- Projector with Python IDE
- Bilingual Python keyword card
- Computers for practical work (if available)
- WORKSHEET 13B
- "Bug Hunt" code snippets (on slides)

--- WORKSHEET 13A ---

WORKSHEET 13A: Algorithms — Searching, Sorting & Trace Tables

Student Name: _____ Date: _____

SECTION A: CONCEPTS

Q1. Explain the difference between decomposition and abstraction in problem solving. Give an example of each. [4]

Q2. a) Describe how a linear search works. [2]

b) State the time complexity of linear search using Big-O notation. [1]

c) When is linear search more suitable than binary search? [1]

Q3. a) Describe how a binary search works. [3]

b) State the time complexity of binary search using Big-O notation. [1]

c) What condition must the data satisfy for binary search to work? [1]

SECTION B: BUBBLE SORT & TRACE TABLES

Q4. Trace the bubble sort algorithm on the following list. Show the state of the list after each pass.

List: [9, 2, 7, 4]

Pass 1: _____ [2]

Pass 2: _____ [2]

Pass 3: _____ [2]

How many comparisons were made in total? _____ [1]

Q5. The following pseudocode performs a linear search. Complete the trace table.

target $\leftarrow$ 7

found $\leftarrow$ FALSE

i $\leftarrow$ 1

numbers $\leftarrow$ [4, 7, 2, 9, 7]

WHILE i $\leq$ 5 AND found = FALSE DO

 IF numbers[i] = target THEN

 found $\leftarrow$ TRUE

 ELSE

 i $\leftarrow$ i + 1

 ENDIF

ENDWHILE

IF found = TRUE THEN

 OUTPUT "Found at position", i

ELSE

 OUTPUT "Not found"

ENDIF

| Pass | i | numbers[i] | found | Action |

What is the output? _____ [1]

Q6. Write pseudocode for a binary search algorithm. [6]

Q7. A programmer needs to search for a customer's name in a list of 10,000 customer records. The list is already sorted alphabetically.

a) Which search algorithm should they use? [1]

b) Justify your answer. [2]

--- WORKSHEET 13B ---

WORKSHEET 13B: Python Programming Review

Student Name: _____ Date: _____

SECTION A: CODE READING (Predict the Output)

Q1. What is the output of this code?


```
name = "Alice"
age = 15
print("Name:", name)
print("Age:", age + 1)
```

Output:

_____ [2]

Q2. What is the output of this code?

```
total = 0
for i in range(1, 6):
    total = total + i
print(total)
```

Output: _____ [2]

Q3. What is the output of this code?

```
x = 10
while x > 0:
    print(x)
    x = x - 3
```

Output:

_____ [2]

Q4. What is the output of this code?

```
score = 75
if score >= 90:
    grade = "A"
elif score >= 80:
    grade = "B"
elif score >= 70:
    grade = "C"
else:
    grade = "D"
print(grade)
```

Output: _____ [1]

SECTION B: CODE WRITING

Q5. Write Python code that asks the user to enter their name and age, then prints a greeting message: "Hello [name], you will be [age+1] next year."

[4]

Q6. Write Python code that asks the user for a number and prints whether it

is positive, negative, or zero.

[4]

Q7. Write Python code that uses a for loop to print the 7 times table from 7×1 to 7×12 , each on a new line.

[3]

Q8. Write Python code that asks the user for 10 numbers and calculates the average. Use a for loop.

[5]

SECTION C: DEBUGGING

Q9. This code should print the factorial of 5 (120). Find and fix the errors.

There are 3 errors.

```
factorial = 0
```

```
for i in range(1, 6)
```

```
    factorial = factorial * i
```

```
print(factorial)
```

Error 1: _____ [1]

Error 2: _____ [1]

Error 3: _____ [1]

Corrected code:

_____ [3]

Q10. This code should find the smallest number in a list. Find and fix the error.

```
numbers = [45, 12, 78, 5, 33]
```

```
smallest = 0
```

```
for num in numbers:
```

```
    if num < smallest:
```

```
        smallest = num
```

```
print(smallest)
```

What does this code actually output? _____ [1]

What should smallest be initialised to? _____ [1]

--- ANSWER KEY (WEEK 13) ---

WORKSHEET 13A — ANSWER KEY

Q1. Decomposition is breaking a complex problem into smaller, more manageable sub-problems (1). Example: creating a game by splitting it into graphics, sound, scoring, and controls (1).

Abstraction is removing unnecessary detail to focus on the essential information (1). Example: a map shows roads but not individual buildings;

a login form hides the encryption details (1).

Q2. a) Linear search checks each item in a list one by one, starting from the first item, until the target is found or the end is reached (1+1).

b) $O(n)$ (1)

c) When the data is unsorted / when the dataset is small (1)

Q3. a) Binary search works on sorted data by repeatedly dividing the search interval in half (1). It compares the target to the middle element; if not found, it searches the left or right half (1). This continues until the target is found or the interval is empty (1).

b) $O(\log n)$ (1)

c) The data must be sorted (1).

Q4. Pass 1: Compare $9 > 2$ swap $\rightarrow [2, 9, 7, 4]$; $9 > 7$ swap $\rightarrow [2, 7, 9, 4]$; $9 > 4$ swap $\rightarrow [2, 7, 4, 9]$ (2)

Pass 2: $2 < 7$ no swap; $7 > 4$ swap $\rightarrow [2, 4, 7, 9]$; $7 < 9$ no swap (2)

Pass 3: $2 < 4$ no swap; $4 < 7$ no swap; already sorted (2)

Total comparisons: 6 ($3+2+1$) (1)

Q5.

Pass	i	numbers[i]	found	Action
1	1	4	FALSE	$4 \neq 7$, increment i
2	2	7	TRUE	$7 = 7$, set found TRUE

(Each row 1 mark, max 2 — accept equivalent)

Output: "Found at position 2" (1)

Q6. Mark scheme (6 marks):

low $\leftarrow$ 1

high $\leftarrow$ LENGTH(list)

found $\leftarrow$ FALSE

WHILE low $\leq$ high AND found = FALSE DO

mid $\leftarrow$ (low + high) DIV 2

IF list[mid] = target THEN

found $\leftarrow$ TRUE

ELSE IF list[mid] < target THEN

low $\leftarrow$ mid + 1

ELSE

high $\leftarrow$ mid - 1

ENDIF

ENDWHILE

(1 mark per correct significant line, max 6)

Key elements: initialise pointers, loop condition, calculate mid, compare, adjust pointers, output result.

Q7. a) Binary search (1)

b) The list is already sorted (1) and binary search is much faster for large datasets — $O(\log n)$ vs $O(n)$ (1). For 10,000 records, binary search needs at most ~14 comparisons vs up to 10,000 for linear.

WORKSHEET 13B — ANSWER KEY

Q1. Name: Alice (1)

Age: 16 (1)

Q2. 15 (2) — sum of 1+2+3+4+5

Q3. 10 (1)

7 (1)

4 (1)

1 (1)

(Give 2 marks if all correct, 1 if pattern understood but minor error)

Q4. C (1)

Q5. Mark scheme (4 marks):

```
name = input("Enter your name: ") (1)
```

```
age = int(input("Enter your age: ")) (1)
```

```
next_age = age + 1 (1, or done inline)
```

```
print("Hello", name, ", you will be", next_age, "next year.")
```

```
OR: print(f"Hello {name}, you will be {age+1} next year.") (1)
```

Accept equivalent approaches. Deduct 1 if no `int()` cast on age.

Q6. Mark scheme (4 marks):

```
num = float(input("Enter a number: ")) (1)
```

```
if num > 0: (1)
```

```
    print("Positive")
```

```
elif num < 0: (1)
```

```
    print("Negative")
```

```
else:
```

```
    print("Zero") (1)
```

Accept: using just `if/else` with nested `if` for zero check.

Q7. Mark scheme (3 marks):

```
for i in range(1, 13): (1)
```

```
    result = 7 * i (1, or inline)
```

```
    print("7 x", i, "=", result) (1)
```

OR: `for i in range(1, 13): print(f"7 x {i} = {7*i}")`

Accept any clear formatting.

Q8. Mark scheme (5 marks):

`total = 0` (1)

`for i in range(10):` (1)

`num = float(input("Enter a number: "))` (1)

`total = total + num` (1)

`average = total / 10` (1)

`print("Average:", average)` (1)

(Max 5 — award best 5)

Q9. Error 1: factorial should be initialised to 1, not 0 (1)

Error 2: missing colon after `range(1, 6)` (1)

Error 3: loop body should multiply factorial by i, but starting at 0 means result stays 0 (1) — this is fixed by error 1 correction.

Corrected code (3 marks):

`factorial = 1`

`for i in range(1, 6):`

`factorial = factorial * i`

`print(factorial)`

Q10. The code outputs: 0 (1) — because 0 is less than all numbers initially,

but the condition `num < smallest` is never true for positive numbers.

`smallest` should be initialised to: `numbers[0]` (or 45, or a very large number) (1)

Corrected code:

`numbers = [45, 12, 78, 5, 33]`

`smallest = numbers[0]`

`for num in numbers:`

`if num < smallest:`

`smallest = num`

`print(smallest)`

WEEK 14: PAPER 2 REVIEW — DATA STRUCTURES & FILE HANDLING

LESSON 14.1 — Arrays, String Handling & Data Structures

Topic: 1D and 2D arrays, string operations, list manipulation

Duration: 40 minutes

Lesson Type: Consolidation / Coding Practice

Syllabus Ref: 2.2 (CIE 0478)

Learning Objective:

By the end of this lesson, students will be able to create and manipulate 1D and 2D arrays, perform string operations, and solve problems using these data structures.

Key Vocabulary:

array / list	— 数组/列表 shù zǔ / liè biǎo
1D array	— 一维数组 yī wéi shù zǔ
2D array	— 二维数组 èr wéi shù zǔ
index	— 索引 suǒ yǐn
element	— 元素 yuán sù
string	— 字符串 zì fú chuàn
slicing	— 切片 qiē piàn
concatenation	— 连接 lián jiē
length	— 长度 cháng dù
traversal	— 遍历 biàn lì
ASCII	— 美国信息交换标准代码 (ASCII)

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"Given: fruits = ['apple', 'banana', 'cherry', 'date']

What is: fruits[1]? fruits[-1]? fruits[1:3]? len(fruits)?"

Answers: 'banana', 'date', ['banana', 'cherry'], 4

3–12 min | DIRECT INSTRUCTION

Topic 1: 1D ARRAYS / LISTS (2 min)

Python lists = 1D arrays.

- Creation: nums = [10, 20, 30, 40, 50]
- Access: nums[0] → 10 (first item, index starts at 0)
- nums[4] → 50 (5th item, index 4)
- nums[-1] → 50 (last item)
- Length: len(nums) → 5
- Modification: nums[2] = 99
- Append: nums.append(60)
- Insert: nums.insert(1, 15) — inserts 15 at index 1
- Delete: del nums[0] or nums.pop(2)

Traversal options:

```
for i in range(len(nums)): # by index  
    print(nums[i])
```

```
for num in nums:      # by element
    print(num)
```

Common operations:

- Find maximum: track largest seen so far
- Find sum: accumulator pattern
- Count occurrences: counter pattern
- Search: check each element

Topic 2: 2D ARRAYS / NESTED LISTS (3 min)

```
grid = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
]
```

- Access: `grid[1][2]` → 6 (row 1, column 2)
- `grid[row][col]` — row first, then column
- Rows: `len(grid)` → 3
- Columns in row 0: `len(grid[0])` → 3

Traversing 2D arrays:

All elements (row by row)

```
for row in range(len(grid)):
    for col in range(len(grid[0])):
        print(grid[row][col])
```

Row sums

```
for row in range(len(grid)):
    row_sum = 0
    for col in range(len(grid[0])):
        row_sum = row_sum + grid[row][col]
    print("Row", row, "sum:", row_sum)
```

Column sums (iterate columns first)

```
for col in range(len(grid[0])):
    col_sum = 0
    for row in range(len(grid)):
        col_sum = col_sum + grid[row][col]
    print("Col", col, "sum:", col_sum)
```

Topic 3: STRING HANDLING (3 min)

- Indexing: `text = "Hello"; text[0]` → 'H'; `text[-1]` → 'o'
- Slicing: `text[1:4]` → 'ell'; `text[:3]` → 'Hel'; `text[2:]` → 'llo'
- Length: `len(text)` → 5

- Concatenation: "Hello" + " " + "World" → "Hello World"
- Upper/lower: text.upper() → "HELLO"; text.lower() → "hello"
- Find: text.find('l') → 2 (first occurrence, -1 if not found)
- ASCII: ord('A') → 65; chr(65) → 'A'
- Substring check: 'ell' in text → True
- Common methods:
 - .strip() — removes whitespace from ends
 - .replace('old', 'new')
 - .split(' ') — splits into list
 - .startswith('He') → True

Topic 4: COMMON EXAM PATTERNS (2 min)

- Finding max/min in a list
- Counting items meeting a condition
- Searching for a value
- Summing rows/columns in 2D arrays
- String validation (length, characters, format)
- Converting between strings and lists

12–25 min | GUIDED PRACTICE

"Calculate the sum of each row in this 2D array:"

```
data = [[5, 3, 1], [2, 8, 4], [7, 0, 9]]
```

Work through on board:

Row 0: 5+3+1 = 9

Row 1: 2+8+4 = 14

Row 2: 7+0+9 = 16

Then write the Python code together (see Topic 2 above).

"What is the output of:"

```
word = "Computer"
```

```
print(word[0:3])    # "Com"
```

```
print(word[-3:])    # "ter"
```

```
print(len(word))    # 8
```

```
print(word.find('p')) # 3
```

25–35 min | INDEPENDENT PRACTICE

Students complete WORKSHEET 14A.

35–40 min | PLENARY + HOMEWORK

Plenary: Code challenge — "Write one line of Python to reverse a string."

Answer: text[::-1]

Homework: Complete WORKSHEET 14A. Create a 3×3 times table using a 2D array in Python.

EAL Support:

- Bilingual data structure vocabulary card
- Visual diagram of 2D array with row/col labels
- Colour-coded index references in code examples
- Pre-completed code templates for common patterns
- String method reference table with examples

Differentiation:

Emerging: Pre-completed code with blanks; supported by TA; focus on 1D arrays and basic string operations (Q1–Q4); visual grid for 2D arrays

Developing: Complete worksheet independently; write row/column sum code; perform string slicing and concatenation

Secure: Complete all questions including stretch (Q9–Q10); write efficient 2D array algorithms; explain difference between by-index and by-element traversal; peer-teach; write column-sum and diagonal-sum code

Resources Needed:

- Projector with Python IDE
- Bilingual keyword sheets
- WORKSHEET 14A
- 2D array visual diagram

--- LESSON 14.2 ---

LESSON 14.2 — Functions, File Handling & Validation

Topic: Functions, scope, file read/write/append, CSV, validation checks

Duration: 40 minutes

Lesson Type: Consolidation / Coding Practice

Syllabus Ref: 2.2 (CIE 0478)

Learning Objective:

By the end of this lesson, students will be able to define and call functions with parameters, read and write files including CSV, and implement all validation checks.

Key Vocabulary:

function	— 函数 hán shù
parameter	— 参数 cān shù
argument	— 参数值 cān shù zhí
return value	— 返回值 fǎn huí zhí
scope	— 作用域 zuò yòng yù
local variable	— 局部变量 jú bù biàn liàng
global variable	— 全局变量 quán jú biàn liàng
file handling	— 文件处理 wén jiàn chǔ lǐ

read	— 读取 dú qǔ
write	— 写入 xiě rù
append	— 追加 zhuī jiā
CSV	— 逗号分隔值 dòu hào fēn gé zhí
validation	— 验证 yàn zhèng
presence check	— 存在性检查 cún zài xìng jiǎn chá
length check	— 长度检查 cháng dù jiǎn chá
type check	— 类型检查 lèi xíng jiǎn chá
range check	— 范围检查 fàn wéi jiǎn chá
format check	— 格式检查 gé shì jiǎn chá

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"Name the five types of validation check and give an example of each."

Students write from memory. Check after 2 minutes.

3–12 min | DIRECT INSTRUCTION

Topic 1: FUNCTIONS (3 min)

- Function: named block of reusable code
- Why use functions: code reuse, organisation, easier debugging, team work

Defining:

```
def greet(name):
    message = "Hello " + name
    return message
```

Calling:

```
result = greet("Alice")
print(result) # Hello Alice
```

Parameters vs arguments:

- Parameter: variable in function definition (name)
- Argument: actual value passed when calling ("Alice")

Return values:

- Functions can return one or more values (Python: return a, b)
- Functions without return statement return None

Scope:

- Local variable: defined inside function, only accessible there
- Global variable: defined outside function, accessible everywhere
- Best practice: use parameters and return values, avoid global variables

Topic 2: FILE HANDLING (3 min)

Three modes:

- 'r' = read (file must exist)
- 'w' = write (creates new or overwrites existing)
- 'a' = append (adds to end of existing file)

Reading:

```
file = open("data.txt", "r")
content = file.read()    # entire file as one string
# OR
lines = file.readlines() # list of lines
# OR
for line in file:        # line by line
    print(line.strip())
file.close()
```

Better: with statement (auto-closes):

```
with open("data.txt", "r") as file:
    for line in file:
        print(line.strip())
```

Writing:

```
with open("output.txt", "w") as file:
    file.write("Hello\n")
    file.write("World\n")
```

Appending:

```
with open("log.txt", "a") as file:
    file.write("New entry\n")
```

CSV handling:

```
with open("data.csv", "r") as file:
    for line in file:
        fields = line.strip().split(",")
        print(fields[0], fields[1])
```

Writing CSV:

```
with open("output.csv", "w") as file:
    file.write("Name,Age,Score\n")
    file.write("Alice,15,85\n")
```

Topic 3: VALIDATION CHECKS (4 min)

58. Presence check: ensure data has been entered (not empty/blank)

```
if len(name) == 0: print("Name required")
```

59. Length check: ensure data is correct length

```
if len(password) < 8: print("Password too short")
```

60. Type check: ensure data is correct data type

```
try: age = int(age_input)
```

```
except: print("Age must be a number")
```

61. Range check: ensure data is within acceptable range

```
if score < 0 or score > 100: print("Score must be 0–100")
```

62. Format check: ensure data matches expected pattern

```
if "@" not in email: print("Invalid email")
```

```
# Or regex for more complex patterns
```

Key exam point: validation checks data is SENSIBLE/REASONABLE, not that it is CORRECT. Only verification (double entry) can check correctness.

12–25 min | GUIDED PRACTICE

"Write a function that takes a list of numbers and returns the average."

Build together:

```
def calculate_average(numbers):
```

```
    if len(numbers) == 0:
```

```
        return 0
```

```
    total = 0
```

```
    for num in numbers:
```

```
        total = total + num
```

```
    return total / len(numbers)
```

"Write code that reads a CSV file of student names and scores, and prints

only those with scores above 80."

```
with open("students.csv", "r") as file:
```

```
    for line in file:
```

```
        data = line.strip().split(",")
```

```
        name = data[0]
```

```
        score = int(data[1])
```

```
        if score > 80:
```

```
            print(name, score)
```

25–35 min | INDEPENDENT PRACTICE

Students complete WORKSHEET 14B.

35–40 min | PLENARY + HOMEWORK

Plenary: "Validation Recap" — quick-fire: I describe a scenario, you name the check.

63. "A form rejects an empty phone number field" → Presence

64. "A password must be at least 8 characters" → Length

- 65. "Age must be a whole number" → Type
- 66. "Age must be between 0 and 120" → Range
- 67. "Email must contain @ and a domain" → Format

Homework: Complete WORKSHEET 14B. Write a program that creates a simple login system using file storage for usernames and passwords.

EAL Support:

- Bilingual file handling and validation vocabulary
- Visual diagram of file open-read-write-close cycle
- Validation checklist card (5 checks with Python examples)
- Scope diagram showing local vs global variables
- Colour-coded function syntax examples

Differentiation:

Emerging: Code skeletons with extensive comments; supported by TA; complete Q1-Q4; validation matching exercise

Developing: Complete worksheet independently; write functions with parameters and return values; implement 3+ validation checks; read and write files

Secure: Complete all questions; write complete programs combining functions, files, and validation; explain scope with examples; peer-teach; implement all 5 validation checks in a single program; use CSV handling

Resources Needed:

- Projector with Python IDE
- Bilingual keyword sheets
- WORKSHEET 14B
- Validation checklist card
- Sample CSV file for demonstrations

--- WORKSHEET 14A ---

WORKSHEET 14A: Arrays, String Handling & Data Structures

Student Name: _____ Date: _____

SECTION A: 1D ARRAYS / LISTS

Q1. Given: scores = [85, 92, 78, 65, 88]

- a) What is scores[2]? _____ [1]
- b) What is scores[-1]? _____ [1]
- c) What is len(scores)? _____ [1]
- d) Write one line of code to change scores[3] to 70. _____ [1]
- e) Write one line of code to add 90 to the end of the list. _____ [1]

Q2. Write Python code to find the largest number in this list.

numbers = [34, 67, 12, 89, 45, 91, 23]

Your code should work for any list of numbers.

[4]

Q3. Write Python code to count how many times the number 7 appears in a list.

```
values = [3, 7, 2, 7, 8, 7, 1]
```

[3]

SECTION B: 2D ARRAYS

Q4. Given this 2D array:

```
grid = [  
    [4, 8, 2],  
    [1, 5, 9],  
    [7, 3, 6]  
]
```

a) What is grid[1][2]? _____ [1]

b) What is grid[0][0]? _____ [1]

c) What does len(grid) return? _____ [1]

d) What does len(grid[0]) return? _____ [1]

Q5. Write Python code to calculate and print the sum of each column in the grid above.

[5]

Q6. Write Python code to find the largest value in the 2D array.

[4]

SECTION C: STRING HANDLING

Q7. Given: text = "Programming"

a) What is text[0:4]? _____ [1]

b) What is text[-3:]? _____ [1]

c) What is len(text)? _____ [1]

d) What does text.find('g') return? _____ [1]

e) What is the output of text.upper()? _____ [1]

Q8. Write Python code that asks the user for their first name and surname, then prints their initials.

Example: "John Smith" → "J.S."

[4]

Q9. Write Python code that checks if a string is a palindrome (reads the same forwards and backwards). Ignore case.

Example: "Radar" → True, "Hello" → False

[4]

WORKSHEET 14B: Functions, File Handling & Validation

Student Name: _____ Date: _____

SECTION A: FUNCTIONS

Q1. a) What is the purpose of a function in programming? [2]

b) What is the difference between a parameter and an argument? [2]

Q2. This function has an error. Find and fix it.

```
def calculate_area(length, width)
    area = length * width
    print(area)
result = calculate_area(5, 3)
print(result)
```

Error: _____ [1]

Fix:

_____ [2]

What will the fixed code output?

_____ [2]

Q3. Write a function called `is_even` that takes a number and returns True if the number is even, False otherwise. Include a docstring.

[4]

Q4. Write a function called `find_max` that takes a list of numbers and returns the largest number. Do NOT use the built-in `max()` function.

[5]

SECTION B: FILE HANDLING

Q5. Write Python code to read a file called "names.txt" and print each line.

[3]

Q6. A file "scores.csv" contains data in this format:

Alice,85

Bob,92

Charlie,78

Write Python code to read this file and print the name and score for each student on separate lines.

[4]

Q7. Write Python code to create a new file called "report.txt" and write the following lines to it:

"Student Report"

"End of report"

[3]

SECTION C: VALIDATION

Q8. For each scenario, name the most appropriate validation check and give a brief description.

a) A form checks that a username has been entered. [2]

Check: _____

Description: _____

b) A password must be at least 8 characters long. [2]

Check: _____

Description: _____

c) Age entered must be a whole number. [2]

Check: _____

Description: _____

d) Test score must be between 0 and 100. [2]

Check: _____

Description: _____

e) Email address must contain @ and a domain name. [2]

Check: _____

Description: _____

Q9. Write Python code that asks the user to enter their age. Your code must include a type check and a range check. If the age is invalid, print an appropriate error message. If valid, print "Age accepted."

[5]

Q10. Explain the difference between validation and verification. [2]

--- ANSWER KEY (WEEK 14) ---

WORKSHEET 14A — ANSWER KEY

Q1. a) 78 (1) b) 88 (1) c) 5 (1)

d) `scores[3] = 70` (1)

e) `scores.append(90)` (1)

Q2. Mark scheme (4 marks):

`largest = numbers[0]` (1)

`for num in numbers:` (1)

`if num > largest:` (1)

`largest = num` (1)

`print(largest)` (1)

(Max 4 — award best 4)

Accept: initialise with 0, but must justify. Using numbers[0] is safest.

Q3. Mark scheme (3 marks):

```
count = 0 (1)
for num in values: (1)
    if num == 7: (1)
        count = count + 1 (1)
print(count) (1)
(Max 3)
```

Q4. a) 9 (1) b) 4 (1) c) 3 (1) d) 3 (1)

Q5. Mark scheme (5 marks):

```
for col in range(len(grid[0])): (1)
    col_sum = 0 (1)
    for row in range(len(grid)): (1)
        col_sum = col_sum + grid[row][col] (1)
    print("Column", col, "sum:", col_sum) (1)
(Max 5)
```

Alternative: hardcoded column access with explanation — max 3.

Q6. Mark scheme (4 marks):

```
largest = grid[0][0] (1)
for row in range(len(grid)): (1)
    for col in range(len(grid[0])): (1)
        if grid[row][col] > largest: (1)
            largest = grid[row][col] (1)
print(largest) (1)
(Max 4)
```

Q7. a) "Prog" (1) b) "ing" (1) c) 11 (1)

d) 3 (first occurrence of 'g') (1) e) "PROGRAMMING" (1)

Q8. Mark scheme (4 marks):

```
first = input("Enter first name: ") (1)
surname = input("Enter surname: ") (1)
initials = first[0] + "." + surname[0] + "." (1)
print(initials) (1)
Accept: first[0].upper() + "." + surname[0].upper() + "."
Accept: f-string or format alternatives.
```

Q9. Mark scheme (4 marks):

```
text = input("Enter a word: ") (1)
```

```

text = text.lower() (1)
if text == text[::-1]: (1)
    print("True")
else:
    print("False") (1)

```

Accept: loop-based comparison.

text[::-1] alone = 2 marks (clever use of slicing).

WORKSHEET 14B — ANSWER KEY

Q1. a) Functions are reusable blocks of code that perform a specific task (1).

They help organise code, reduce repetition, and make debugging easier (1).

b) A parameter is a variable defined in the function header (1).

An argument is the actual value passed to the function when it is called (1).

Q2. Error: Missing colon after function definition (1)

Fix:

```

def calculate_area(length, width):
    area = length * width
    return area (1 for colon, 1 for return instead of print)

```

Output of fixed code:

15 (1 — from print(result))

None (1 — if not fixed with return; accept either based on fix)

Actually: if print(area): outputs 15 then None (printed from print(result))

If return area: outputs 15 only

Mark scheme: Error (1), Colon fix (1), return statement (1), correct output 15 (1),

no extra None output (1). Max 5 available, question asks for 5 marks.

Award based on student's fix.

Q3. Mark scheme (4 marks):

```

def is_even(number): (1)
    """Returns True if number is even, False otherwise.""" (1 for docstring)
    if number % 2 == 0: (1)
        return True (0.5)
    else:
        return False (0.5)

```

Accept: return number % 2 == 0 (compact, 2 marks for logic)

Q4. Mark scheme (5 marks):

```

def find_max(numbers): (1)
    largest = numbers[0] (1)

```

```

for num in numbers: (1)
    if num > largest: (1)
        largest = num (1)
return largest (1)

```

(Max 5)

Q5. Mark scheme (3 marks):

```

with open("names.txt", "r") as file: (1)
    for line in file: (1)
        print(line.strip()) (1)

```

Accept: file = open/read/close approach.

Q6. Mark scheme (4 marks):

```

with open("scores.csv", "r") as file: (1)
    for line in file: (1)
        data = line.strip().split(",") (1)
        print(data[0], data[1]) (1)

```

Accept: name, score = line.strip().split(",") then print.

Q7. Mark scheme (3 marks):

```

with open("report.txt", "w") as file: (1)
    file.write("Student Report\n") (1)
    file.write("=====\n") (1)
    file.write("End of report\n") (1)

```

(Max 3 — any 3 correctly written lines)

Q8. a) Presence check (1): checks that data has been entered/is not blank (1)

b) Length check (1): checks that data meets minimum/maximum length requirements (1)

c) Type check (1): checks that data is the correct data type (e.g., integer) (1)

d) Range check (1): checks that data falls within an acceptable range (1)

e) Format check (1): checks that data matches an expected pattern or format (1)

Q9. Mark scheme (5 marks):

```

age_input = input("Enter your age: ") (1)
try: (1)
    age = int(age_input) (1)
    if age >= 0 and age <= 120: (1)
        print("Age accepted.")
    else:
        print("Age must be between 0 and 120.") (1)
except:

```

```
print("Age must be a whole number.") (1)
```

(Max 5)

Accept: nested if structure or combined checks in any logical order.

Q10. Validation checks that data is reasonable/sensible/fits the expected format (1).

Verification checks that data is correct/accurate, usually by entering it twice or checking against an original source (1).

WEEK 15: PAPER 2 REVIEW — DATABASES, BOOLEAN LOGIC & SCENARIO QUESTION

LESSON 15.1 — SQL, Database Concepts & Boolean Logic

Topic: SQL commands, database design, normalisation, all logic gates, truth tables

Duration: 40 minutes

Lesson Type: Consolidation / Exam Practice

Syllabus Ref: 2.3, 2.4 (CIE 0478)

Learning Objective:

By the end of this lesson, students will be able to write SQL queries using all required commands, explain database design concepts, construct truth tables for all logic gates, and simplify logic expressions.

Key Vocabulary:

database — 数据库 shù jù kù

table — 表 biǎo

record — 记录 jì lù

field — 字段 zì duàn

primary key — 主键 zhǔ jiàn

foreign key — 外键 wài jiàn

relational database — 关系数据库 guān xì shù jù kù

flat-file database — 平面文件数据库 píng miàn wén jiàn shù jù kù

normalisation — 规范化 guī fàn huà

entity — 实体 shí tǐ

Boolean logic — 布尔逻辑 bù ěr luó jí

logic gate — 逻辑门 luó jí mén

truth table — 真值表 zhēn zhí biǎo

logic expression — 逻辑表达式 luó jí biǎo dá shì

logic circuit — 逻辑电路 luó jí diàn lù

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"Write the SQL query to display all students in Year 11 from a table called Students." Students write in books.

Answer: `SELECT * FROM Students WHERE Year = 11;`

3–12 min | DIRECT INSTRUCTION

Topic 1: SQL COMMANDS (3 min)

Core commands for CIE 0478:

- `SELECT field1, field2 FROM table;` — select specific fields
- `SELECT * FROM table;` — select all fields
- `SELECT * FROM table WHERE condition;` — filter rows
- `SELECT * FROM table WHERE field LIKE 'A%';` — wildcard (% = any chars, _ = one char)
- `SELECT * FROM table WHERE condition1 AND condition2;` — both must be true
- `SELECT * FROM table WHERE condition1 OR condition2;` — at least one true
- `SELECT * FROM table ORDER BY field ASC/DESC;` — sort results
- `INSERT INTO table (field1, field2) VALUES (value1, value2);` — add record
- `DELETE FROM table WHERE condition;` — remove record (ALWAYS use WHERE!)

Common conditions:

- `field = 'value', field > 100, field <> 'X'` (not equal)
- `field BETWEEN 10 AND 20`
- `field LIKE 'S%'` (starts with S)
- `field IN ('A', 'B', 'C')`

Exam tip: SQL keywords are not case-sensitive, but convention is UPPERCASE.

String values must be in quotes. Numbers do not need quotes.

Topic 2: DATABASE CONCEPTS (2 min)

- Flat-file database: all data in one table. Problems: data redundancy, update anomalies, data inconsistency.
- Relational database: data spread across linked tables. Reduces redundancy.
- Primary key: unique identifier for each record in a table.

Rules: unique, never NULL, minimal fields, never changes.

- Foreign key: field in one table that links to primary key in another table.

Creates relationships between tables.

- Entity: a real-world object represented as a table (e.g., Student, Course).
- Entity-Relationship (ER) diagram: shows tables and their relationships.
- Normalisation: organising data to reduce redundancy. Process:
 - 1NF: no repeating groups, atomic values
 - 2NF: 1NF + no partial dependencies
 - 3NF: 2NF + no transitive dependencies
- Benefits of normalisation: reduced redundancy, data integrity, easier to update, more efficient storage.

Topic 3: BOOLEAN LOGIC — ALL 6 GATES (3 min)

NOT gate (inverter): output is opposite of input

A | OUTPUT

0 | 1

1 | 0

AND gate: output is 1 only if ALL inputs are 1

A B | OUTPUT

0 1 | 0

1 0 | 0

OR gate: output is 1 if ANY input is 1

A B | OUTPUT

0 1 | 1

1 0 | 1

NAND gate: AND followed by NOT (opposite of AND)

A B | OUTPUT

0 0 | 1

0 1 | 1

1 0 | 1

1 1 | 0

NOR gate: OR followed by NOT (opposite of OR)

A B | OUTPUT

0 0 | 1

0 1 | 0

1 0 | 0

1 1 | 0

XOR gate: output is 1 if inputs are DIFFERENT

A B | OUTPUT

0 1 | 1

1 0 | 1

1 1 | 0

Topic 4: LOGIC EXPRESSIONS & CIRCUITS (2 min)

- Logic expression uses AND, OR, NOT: e.g., $OUTPUT = A \text{ AND } (B \text{ OR NOT } C)$
- To evaluate: work from inner brackets outward, apply NOT first, then AND, then OR
- Logic circuit: diagram showing gates connected together
- Each gate has specific symbol; lines connect outputs to inputs
- Exam tip: always construct truth table systematically — list all input combinations

first (2^n rows for n inputs)

12–25 min | GUIDED PRACTICE

SQL practice: Given a Students table (StudentID, FirstName, LastName, Year, House, Score)

Q1: "Write SQL to display first names and scores of all Year 10 students."

→ SELECT FirstName, Score FROM Students WHERE Year = 10;

Q2: "Write SQL to find all students in House 'Red' with Score > 80."

→ SELECT * FROM Students WHERE House = 'Red' AND Score > 80;

Q3: "Write SQL to list students whose last name starts with 'S'."

→ SELECT * FROM Students WHERE LastName LIKE 'S%';

Boolean practice: Construct truth table for A AND (B OR NOT C)

| A | B | C | NOT C | B OR NOT C | A AND (B OR NOT C) |

| 0 | 0 | 0 | 1 | 1 | 0 |

| 0 | 0 | 1 | 0 | 0 | 0 |

| 0 | 1 | 0 | 1 | 1 | 0 |

| 0 | 1 | 1 | 0 | 1 | 0 |

| 1 | 0 | 0 | 1 | 1 | 1 |

| 1 | 0 | 1 | 0 | 0 | 0 |

| 1 | 1 | 0 | 1 | 1 | 1 |

| 1 | 1 | 1 | 0 | 1 | 1 |

25–35 min | INDEPENDENT PRACTICE

Students complete WORKSHEET 15A.

35–40 min | PLENARY + HOMEWORK

Plenary: Boolean rapid-fire — teacher calls gate + inputs, students shout output.

"NAND: 1, 0" → "1"; "XOR: 1, 1" → "0"; "NOR: 0, 0" → "1"

Homework: Complete WORKSHEET 15A. Practise writing SQL queries on

<https://sqlbolt.com> (optional).

EAL Support:

- Bilingual SQL keyword reference card
- Truth table templates with all gate symbols
- Colour-coded logic circuit diagram
- SQL sentence frames: "SELECT [fields] FROM [table] WHERE [condition]"
- Visual ER diagram example with labels in both languages

Differentiation:

Emerging: Pre-completed SQL templates with blanks; supported by TA; complete Q1–Q4;

truth table with some outputs filled; logic gate matching cards

Developing: Write SQL queries independently; complete truth tables from scratch;

explain database concepts in own words; draw simple logic circuits

Secure: Complex SQL with AND/OR combinations; evaluate logic expressions with multiple gates; design normalised database schemas; peer-teach; complete all stretch questions; simplify logic expressions using Boolean algebra

Resources Needed:

- Projector
- SQL keyword reference card
- Truth table templates
- Logic gate symbol reference
- WORKSHEET 15A
- Optional: SQL practice website

--- LESSON 15.2 ---

LESSON 15.2 — The 15-Mark Scenario Question: Strategy & Practice

Topic: Approaching the 15-mark programming scenario question

Duration: 40 minutes

Lesson Type: Exam Technique / Extended Practice

Syllabus Ref: 2.2 (CIE 0478)

Learning Objective:

By the end of this lesson, students will know how to approach the 15-mark scenario question systematically and will have practised with a past-style question.

Key Vocabulary:

scenario question — 情景题 qíng jǐng tí

strategy — 策略 cè lüè

planning — 计划 jì huà

pseudocode — 伪代码 wěi dài mǎ

marks per minute — 每分钟得分 měi fēn zhōng dé fēn

common pitfall — 常见陷阱 cháng jiàn xiàn jǐng

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"You have 30 minutes for a 15-mark programming question. How many minutes per mark?" Answer: 2 minutes per mark.

"What should you do in the first 5 minutes?" Brainstorm ideas.

3–12 min | DIRECT INSTRUCTION: THE STRATEGY

THE 5-STEP APPROACH TO THE 15-MARK SCENARIO:

STEP 1: READ & UNDERSTAND (3 minutes)

- Read the scenario TWICE.
- Underline: inputs required, outputs required, processing needed, any

validation rules, file operations.

- Identify variables and data structures needed.
- Note: what is the program supposed to DO?

STEP 2: PLAN (4 minutes)

- Write a brief plan in plain English or as comments:

1. Get input from user

2. Validate input

3. Read data from file

4. Process data

5. Output results

- List variables with their purposes.
- Identify any loops or conditions needed.
- Note: Cambridge marks code that WORKS, not beautiful plans. But a plan

helps you write working code faster.

STEP 3: WRITE CODE (15 minutes)

- Write in pseudocode OR Python — both accepted.
- If using Python: correct indentation is ESSENTIAL.
- Use meaningful variable names.
- Include comments explaining what each section does.
- Write the MAIN STRUCTURE first, then fill in details.
- Make sure every requirement from the scenario is addressed.

STEP 4: TEST (5 minutes)

- Trace through your code with sample data.
- Check: does it produce the correct output?
- Check edge cases: empty input, maximum values, boundary conditions.
- If you find an error — FIX IT! Even small fixes gain marks.

STEP 5: REVIEW (3 minutes)

- Check you haven't missed any requirements.
- Ensure variable names are consistent throughout.
- Add any missing comments.

COMMON PITFALLS (what students lose marks on):

Not reading the whole question before starting

Forgetting to validate inputs

Using wrong data types (e.g., comparing string to number)

Off-by-one errors in loops

Not initialising variables before use

Writing code that doesn't match the scenario requirements

Mixing pseudocode and Python inconsistently

Not indenting Python code correctly

Spending too long on planning and running out of writing time

Leaving the question blank — ALWAYS attempt something!

MARK ALLOCATION (typical 15-mark question):

- Variable declaration/initialisation: 1–2 marks
- Input with appropriate prompts: 1–2 marks
- Validation checks: 2–3 marks
- Main processing logic: 4–6 marks
- Output/display: 1–2 marks
- Correct use of loops/conditions: 2–3 marks
- File operations (if required): 2–3 marks
- Code structure and readability: 1–2 marks

12–25 min | GUIDED PRACTICE

Walk through a sample scenario together:

"A teacher needs a program to record test scores for students. The program should:

- Ask for the student's name
- Ask for the test score (0–100)
- Validate the score is within range
- Store the name and score in a file called results.txt
- Ask if there is another student to enter (Y/N)
- If Y, repeat; if N, display how many students were entered"

Plan together:

68. Variables: name (string), score (integer), count (integer), another (string)

69. Open file for writing

70. REPEAT:

- a. Input name
- b. Input score
- c. Validate score (0–100)
- d. Write to file
- e. Increment count
- f. Ask if another

UNTIL another = "N"

71. Display count

72. Close file

25–40 min | INDEPENDENT PRACTICE

Students complete WORKSHEET 15B — the full scenario question (30 min timed).

Teacher circulates; provides encouragement and minimal hints.

Remind students to use the 5-step approach.

Time warnings at 15 min and 5 min remaining.

Homework: Review your scenario answer. Identify at least 3 ways you could improve it. Write an improved version.

EAL Support:

- Strategy checklist card in both languages
- 5-step approach poster for classroom wall
- Scenario question vocabulary list
- Sentence frames for planning comments
- Extra reading time for scenario text

Differentiation:

Emerging: Shorter scenario (8 marks); code skeleton provided with extensive comments; supported by TA; focus on completing input/output and one validation check

Developing: Complete standard 15-mark scenario; use planning template; attempt all requirements; use pseudocode or Python

Secure: Complete scenario efficiently; use advanced features (functions, file handling, multiple validation checks); complete with time to review and test; peer-assess another student's answer using mark scheme

Resources Needed:

- 5-step approach poster (printed large)
- Planning template (blank copies)
- WORKSHEET 15B (scenario question)
- Mark scheme for scenario question (teacher copy)
- Timer
- Common pitfalls poster

--- WORKSHEET 15A ---

WORKSHEET 15A: SQL, Databases & Boolean Logic

Student Name: _____ Date: _____

SECTION A: SQL

A school database has a table called Students with these fields:

StudentID, FirstName, LastName, YearGroup, House, ExamScore

Write SQL queries for the following:

Q1. Display all fields for all students. [1]

Q2. Display the first name and last name of all students in Year 10. [2]

Q3. Display all students in House 'Blue' with an exam score above 75. [2]

Q4. Display all students whose last name starts with 'M'. [2]

Q5. Display all students sorted by ExamScore in descending order. [2]

Q6. Add a new student: StudentID=106, FirstName='Li', LastName='Wei',
YearGroup=10, House='Red', ExamScore=82. [2]

Q7. Delete the student with StudentID 103. [1]

Q8. Display students in Year 10 OR Year 11 who scored at least 60. [3]

SECTION B: DATABASE CONCEPTS

Q9. a) What is a primary key? [2]

b) State two properties of a good primary key. [2]

Q10. Explain the difference between a flat-file database and a relational database. [3]

Q11. Explain why normalising a database is important. [2]

Q12. Two tables are linked: Orders (OrderID, CustomerID, Date, Total) and Customers (CustomerID, FirstName, LastName, Email).

a) What is the primary key of the Customers table? [1]

b) Which field in the Orders table is a foreign key? [1]

c) Explain the purpose of this foreign key. [2]

SECTION C: BOOLEAN LOGIC

Q13. Draw the truth table for a NAND gate. [4]

A	B	OUTPUT
---	---	--------

Q14. Draw the truth table for an XOR gate. [4]

A	B	OUTPUT
---	---	--------

Q15. Complete the truth table for this logic expression:

OUTPUT = (A AND B) OR NOT C

A	B	C	A AND B	NOT C	OUTPUT
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			

Q16. A security system uses this logic:

Alarm = (DoorOpen AND MotionDetected) OR NOT SecurityEnabled

a) Complete the truth table: [4]

Door	Motion	Security	Alarm
Open	Detected	Enabled	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	

| 1 | 1 | 0 | |

b) In which scenario is the alarm OFF when it should ideally be ON?

Explain why this might be a security problem. [2]

Q17. Draw the logic circuit for: $\text{OUTPUT} = A \text{ AND } (B \text{ OR NOT } C)$

Label all gates clearly. [4]

[Use this space to draw]

--- WORKSHEET 15B ---

WORKSHEET 15B: The 15-Mark Scenario Question

Student Name: _____ Date: _____

SCENARIO: STUDENT ATTENDANCE SYSTEM

A teacher wants a program to track student attendance for a class.

The program should:

1. Ask the teacher to enter the class name.

73. Ask how many students are in the class (must be between 1 and 35).

Validate this input.

3. For each student:

- Ask for the student's name.
- Ask if the student is present (P) or absent (A).
- Validate that only 'P' or 'A' is entered.

74. Store all attendance data in a file called "attendance.txt" in this format:

ClassName: [class name]

StudentName, Present/Absent

(one line per student)

5. After all students are entered, display:

- Total number of students
- Number present
- Number absent
- Attendance percentage ($\text{present} / \text{total} \times 100$)

75. Ask the teacher if they want to record attendance for another class (Y/N).

If Y, start again. If N, end the program.

Write your solution in pseudocode OR Python.

Use the planning space below before you start coding.

PLANNING SPACE (write your plan here):

YOUR CODE:

--- ANSWER KEY (WEEK 15) ---

WORKSHEET 15A — ANSWER KEY

Q1. SELECT * FROM Students; (1)

Q2. SELECT FirstName, LastName FROM Students WHERE YearGroup = 10; (1+1)

Q3. SELECT * FROM Students WHERE House = 'Blue' AND ExamScore > 75; (1+1)

Q4. SELECT * FROM Students WHERE LastName LIKE 'M%'; (1+1)

Accept: LIKE 'M_____' (with correct length)

Q5. SELECT * FROM Students ORDER BY ExamScore DESC; (1+1)

Deduct 1 if DESC missing (ASC would be wrong for this question).

Q6. INSERT INTO Students (StudentID, FirstName, LastName, YearGroup, House, ExamScore)
VALUES (106, 'Li', 'Wei', 10, 'Red', 82); (1+1)

Accept without field list: INSERT INTO Students VALUES (106, 'Li', 'Wei', 10, 'Red', 82);

Q7. DELETE FROM Students WHERE StudentID = 103; (1)

CRITICAL: must have WHERE clause. Without WHERE = all records deleted (0 marks).

Q8. SELECT * FROM Students WHERE (YearGroup = 10 OR YearGroup = 11) AND ExamScore
>= 60;

(1 for OR, 1 for AND, 1 for correct conditions)

Accept: IN operator: WHERE YearGroup IN (10, 11) AND ExamScore >= 60

Brackets around OR condition essential — deduct 1 if missing.

Q9. a) A primary key is a unique identifier for each record in a database table (1+1).

b) Any two from: unique (no duplicates), never NULL, never changes, minimal
number of fields, automatically generated (1 each, max 2).

Q10. A flat-file database stores all data in a single table (1). A relational database
stores data across multiple linked tables (1), connected by primary and foreign keys (1).
Relational databases reduce data redundancy and improve data integrity.

Q11. Normalisation reduces data redundancy / prevents duplicate data (1). It prevents
update anomalies and improves data integrity/consistency (1).

(Any valid benefit = 1 mark)

Q12. a) CustomerID (1)

b) CustomerID (1)

c) The foreign key links each order to the customer who placed it (1), enabling
the database to join the two tables and retrieve customer details for any order (1).

Q13. NAND truth table (1 mark per correct row, max 4):

A	B	OUTPUT
---	---	--------

0	0	1
---	---	---

0	1	1
---	---	---

	1		0		1	
	1		1		0	

Q14. XOR truth table (1 mark per correct row, max 4):

A B OUTPUT
0 1 1
1 0 1
1 1 0

Q15. (A AND B) OR NOT C truth table (0.5 per cell, max 8):

A B C A AND B NOT C OUTPUT
0 0 0 0 1 1
0 0 1 0 0 0
0 1 0 0 1 1
0 1 1 0 0 0
1 0 0 0 1 1
1 0 1 0 0 0
1 1 0 1 1 1
1 1 1 1 0 1

Q16. a) Truth table (0.5 per cell, max 4):

Door Motion Security Alarm
Open Detected Enabled
0 0 0 1
0 0 1 0
0 1 0 1
0 1 1 0
1 0 0 1
1 0 1 0
1 1 0 1

b) The alarm is OFF (0) when Door=0, Motion=0, Security=1 (row 2) — but also when Door=0, Motion=1, Security=1 (row 4) and Door=1, Motion=0, Security=1 (row 6).

This is a security problem because when the security system is enabled (Security=1), the alarm is OFF even when motion is detected but the door is closed (row 4).

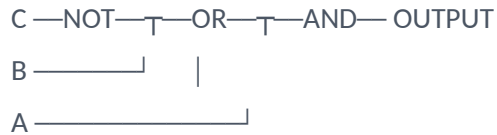
This means an intruder could enter through a window (not the door) and the alarm would not trigger. (1 for identifying scenario + 1 for explanation)

Accept: the NOT SecurityEnabled term means alarm triggers when security is disabled, which is backwards — this is a logic design flaw.

Q17. Logic circuit (1 mark per correct component, max 4):

- NOT gate on C (1)
- OR gate with B and NOT C (1)
- AND gate with A and output of OR (1)
- Correct overall structure (1)

ASCII representation:



WORKSHEET 15B — ANSWER KEY & MARK SCHEME

15-Mark Scenario Question: Student Attendance System

Mark scheme breakdown:

[1 mark] Variable declaration/initialisation:

class_name (string), num_students (integer), student_name (string),
attendance (string), present_count (integer), absent_count (integer),
another_class (string) — at least 3 sensibly named variables = 1 mark

[2 marks] Input class name:

INPUT class_name (1)

Appropriate prompt (1)

[2 marks] Input and validate number of students:

INPUT num_students (1)

WHILE num_students < 1 OR num_students > 35 DO (1)

OUTPUT error message (1)

INPUT num_students (1)

(Validation loop = 2 marks max)

[2 marks] Loop through students:

FOR i ← 1 TO num_students DO (1)

or REPEAT ... UNTIL all students entered (1)

Correct loop structure and iteration count (1)

[2 marks] Input student name and attendance:

INPUT student_name (0.5)

INPUT attendance (0.5)

Prompt for each (0.5+0.5)

[2 marks] Validate attendance:

WHILE attendance <> 'P' AND attendance <> 'A' DO (1)

OUTPUT error message (1)

INPUT attendance (1)

Correct validation of both P and A (1)

[2 marks] Write to file:

OPEN "attendance.txt" FOR WRITE (or APPEND) (1)

Write class name header (1)

Write each student's data on separate line (1)

Correct format (Name, P/A) (1)

Close file (1)

(File operations = 2 marks max)

[2 marks] Count present/absent:

IF attendance = 'P' THEN present_count ← present_count + 1 (1)

ELSE absent_count ← absent_count + 1 (1)

Correct initialisation of counters (1)

[2 marks] Calculate and display statistics:

percentage ← (present_count / num_students) × 100 (1)

Display total, present, absent, percentage (1+1 — all four = 2 marks)

[2 marks] Outer loop for another class:

INPUT another_class (1)

REPEAT entire program UNTIL another_class = 'N' (1)

or WHILE another_class = 'Y' DO (1)

Correct loop structure encompassing entire process (1)

[1 mark] Code structure and readability:

Meaningful variable names (0.5)

Comments explaining sections (0.5)

Correct indentation (if Python) (0.5)

(Max 1)

Total: 18 marks available → cap at 15. Award marks holistically if code structure varies but achieves all requirements.

Sample solution (Python):

Student Attendance System

```
another_class = "Y"
```

```
while another_class == "Y":
```

```
    # Input class name
```

```
    class_name = input("Enter class name: ")
```

```
    # Input and validate number of students
```

```
    num_students = int(input("How many students (1-35): "))
```

```

while num_students < 1 or num_students > 35:
    print("Must be between 1 and 35.")
    num_students = int(input("How many students (1-35): "))
# Open file
with open("attendance.txt", "w") as file:
    file.write("ClassName: " + class_name + "\n")
    present_count = 0
    absent_count = 0
    # Record each student
    for i in range(num_students):
        name = input("Enter student name: ")
        attendance = input("Present (P) or Absent (A): ")
        while attendance != "P" and attendance != "A":
            print("Enter P or A only.")
            attendance = input("Present (P) or Absent (A): ")
        file.write(name + ", " + attendance + "\n")
        if attendance == "P":
            present_count = present_count + 1
        else:
            absent_count = absent_count + 1
    # Display statistics
    print("Total students:", num_students)
    print("Present:", present_count)
    print("Absent:", absent_count)
    percentage = (present_count / num_students) * 100
    print("Attendance percentage:", percentage, "%")
    another_class = input("Another class? (Y/N): ")

```

WEEK 16: MIXED PAPER 2 PRACTICE

LESSON 16.1 — Mixed Paper 2 Practice (Timed Section)

Topic: All Paper 2 topics in timed exam conditions

Duration: 40 minutes

Lesson Type: Exam Practice / Timed Assessment

Syllabus Ref: All Paper 2 topics (CIE 0478)

Learning Objective:

By the end of this lesson, students would have completed a timed section of

mixed Paper 2 questions under exam conditions, practising time management and exam technique.

Key Vocabulary:

timed practice — 限时练习 xiàn shí liàn xí

exam conditions — 考试条件 kǎo shì tiáo jiàn

time management — 时间管理 shí jiān guǎn lǐ

show your working — 展示你的过程 zhǎn shì nǐ de guò chéng

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"In Paper 2, you have 1 hour 30 minutes for 50 marks. How many minutes per mark?"

Answer: 1.8 minutes per mark (about 1 minute 50 seconds per mark).

"For a 4-mark pseudocode question, how long should you spend?" About 7–8 minutes.

3–5 min | EXAM TECHNIQUE BRIEFING

Paper 2 specific tips:

- Pseudocode: use Cambridge conventions, meaningful variable names, comments
- Python: CORRECT INDENTATION is essential (4 spaces per level)
- Trace tables: show ALL columns, work through step by step
- Code reading: trace through mentally or with quick notes
- Code writing: plan first (even briefly), then write
- If using Python, you can use pseudocode instead — both accepted
- Show your working in trace tables — partial marks available
- Check: have you answered EXACTLY what was asked?

Time allocation guide:

- Short questions (1–2 marks): 2–4 minutes each
- Code reading (3–4 marks): 5–7 minutes
- Trace table (4–6 marks): 8–10 minutes
- Code writing (4–6 marks): 8–12 minutes
- Scenario question (15 marks): 25–30 minutes

5–38 min | TIMED PRACTICE

Students complete WORKSHEET 16A — timed 35-minute section.

STRICT exam conditions: no talking, no phones, no notes, no calculators (unless specified).

Teacher circulates silently. Time warnings at 20 min, 10 min, 5 min remaining.

For students who finish early: review answers, check for common errors.

38–40 min | END OF LESSON

"Put your pens down. We will review answers tomorrow."

Collect all work.

Homework: Review all Paper 2 topics. Focus on your weakest areas from

self-assessment checklist (provided).

EAL Support:

- Exam technique card with Paper 2 tips in both languages
- Clarify any vocabulary in questions before timed section begins
- Extra time consideration (discuss with SENCO)
- Allow question clarification during reading time (not during writing time)

Differentiation:

Emerging: Shorter timed section (20 min); fewer questions; can use reference card; supported seating near TA; scaffolded with code skeletons for writing questions

Developing: Complete standard 35-min timed section; attempt all questions

Secure: Complete 35-min section plus additional challenge questions; review and self-check answers; act as "exam buddy" by helping peer review next lesson

Resources Needed:

- WORKSHEET 16A (printed, one per student)
- Timer (visible to all)
- Exam conditions poster
- Spare pens

--- LESSON 16.2 ---

LESSON 16.2 — Paper 2 Review, Common Errors & Exam Technique

Topic: Reviewing Paper 2 answers, identifying common errors, exam technique

Duration: 40 minutes

Lesson Type: Review / Exam Technique

Syllabus Ref: All Paper 2 topics (CIE 0478)

Learning Objective:

By the end of this lesson, students will have reviewed their Paper 2 practice answers, identified common errors, and consolidated exam technique for maximising marks.

Key Vocabulary:

peer assessment	— 同伴评估 tóng bàn píng gū
common error	— 常见错误 cháng jiàn cuò wù
mark scheme	— 评分标准 píng fēn biāo zhǔn
self-assessment	— 自我评估 zì wǒ píng gū

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"What was the hardest question in yesterday's paper? Write the question number and one word to describe why it was hard."

Collect responses on board — identify common themes.

3–25 min | REVIEW ANSWERS

Go through WORKSHEET 16A mark scheme systematically:

For each question:

76. Read the question aloud

77. Ask: "What was the mark scheme looking for?"

78. Show common errors

79. Show model answer

80. Students self-mark in a different colour pen

MOST COMMON PAPER 2 ERRORS to highlight:

81. Pseudocode / Python errors:

- Forgetting colons in Python (if, for, while, def)
- Wrong indentation (mixed tabs/spaces, inconsistent levels)
- Using = instead of == for comparison
- Not initialising variables before loops
- Off-by-one errors (e.g., range(10) gives 0–9, not 1–10)
- Not casting input() to int/float when needed

82. Trace table errors:

- Missing columns
- Stopping too early (not completing all iterations)
- Not updating variables correctly
- Forgetting that array indices start at 0 (in Python) or 1 (in pseudocode)

83. Algorithm errors:

- Not stating what the algorithm does at the end
- Missing the final output line in trace tables
- Wrong loop condition (<= vs <)
- Not handling edge cases (empty list, single item)

84. SQL errors:

- Missing quotes around string values
- Missing WHERE in DELETE (would delete everything!)
- Confusing AND and OR
- Wrong wildcard syntax (% vs *)

25–35 min | PEER ASSESSMENT

Students swap papers with a partner. Check:

- Did they attempt every question?
- Is the indentation correct (for Python)?
- Did they show working in trace tables?
- Are variable names meaningful?

Partners give ONE strength and ONE improvement suggestion.

35–40 min | PLENARY + TARGET SETTING

Plenary: Each student writes ONE specific target for Paper 2 improvement on a sticky note. Stick on "target board" at front.

Examples: "I will check my indentation every time", "I will always trace through my code before finishing", "I will practise SQL wildcards."

Homework: Complete WORKSHEET 16B (additional mixed practice). Focus on the topic of your sticky note target.

EAL Support:

- Mark scheme available in both languages for key questions
- Peer assessment guidance with sentence frames: "One strength is..."
- Visual common errors checklist
- One-to-one discussion of errors with TA

Differentiation:

Emerging: Supported by TA during review; self-mark with guidance; focus on identifying correct answers rather than errors; reteach of weakest topic

Developing: Self-mark independently; identify own errors; set specific target; peer assess using mark scheme

Secure: Self-mark; peer assess; explain errors to others; create improved versions of their answers; lead pair review discussions

Resources Needed:

- Mark scheme (printed or projected)
- Students' completed WORKSHEET 16A (returned)
- Sticky notes
- Target board
- Common errors checklist (projected or printed)
- WORKSHEET 16B (for homework)

--- WORKSHEET 16A ---

WORKSHEET 16A: Mixed Paper 2 Practice — TIMED (35 minutes)

Student Name: _____ Date: _____

Instructions: You have 35 minutes. Answer ALL questions. Write in full sentences for written answers. Use correct indentation for Python code. Show all working.

Total: 45 marks

Q1. The following pseudocode uses a bubble sort algorithm.

```
PROCEDURE BubbleSort(data)
  n ← LENGTH(data)
  FOR i ← 1 TO n-1
    FOR j ← 1 TO n-i
      IF data[j] > data[j+1] THEN
        temp ← data[j]
        data[j] ← data[j+1]
        data[j+1] ← temp
```

```
ENDIF
NEXT j
NEXT i
ENDPROCEDURE
```

- State the time complexity of bubble sort using Big-O notation. [1]
- Describe one way the algorithm above could be made more efficient. [2]
- Trace the algorithm with this data: [5, 2, 8, 1]

Show the state of the array after each pass.

Pass 1: _____ [1]

Pass 2: _____ [1]

Pass 3: _____ [1]

Q2. Write pseudocode OR Python for a linear search algorithm.

Your algorithm should:

- Take a list and a target value as inputs
- Return the position of the target if found, or -1 if not found

[5]

Q3. A database table called Products contains: ProductID, Name, Price, Category, Stock

- Write SQL to display all products with stock less than 10. [2]
- Write SQL to add a new product with ID=50, Name='Mouse', Price=25.99, Category='Electronics', Stock=100. [2]
- Write SQL to increase the price of all products in the 'Electronics' category by 10%. [3]
- Write SQL to display the most expensive product in each category. [3]

Q4. Study this Python code:

```
def mystery(text):
    result = ""
    for char in text:
        if char in "AEIOUaeiou":
            result = result + char.upper()
        else:
            result = result + char
    return result
```

- What is returned by mystery("Hello")? [2]
- What is returned by mystery("Python")? [2]
- Describe what this function does in one sentence. [2]

Q5. Trace the following pseudocode with the given input:

```

num ← 27
count ← 0
WHILE num > 0 DO
  IF num MOD 2 = 0 THEN
    num ← num DIV 2
  ELSE
    num ← num - 1
  ENDIF
  count ← count + 1
ENDWHILE
OUTPUT count

```

Complete the trace table:

Pass	num	num MOD 2	Action	count

What is the final output? _____ [1]

Q6. Write a Python function called `count_words` that takes a string and returns the number of words in the string. Words are separated by spaces.

Example: `count_words("Hello world")` returns 2

[5]

Q7. Complete the truth table for this logic expression:

OUTPUT = (NOT A AND B) OR (A AND NOT B)

A	B	NOT A	NOT B	NOT A AND B	A AND NOT B	OUTPUT
0	1					
1	0					

What logic gate does this expression represent? _____ [1]

Q8. Write pseudocode OR Python that:

- Creates a 1D array of 5 integers input by the user
- Calculates the sum and average
- Counts how many numbers are above the average
- Displays the sum, average, and count

[8]

--- WORKSHEET 16B ---

WORKSHEET 16B: Additional Mixed Paper 2 Practice

Student Name: _____ Date: _____

Q1. Write pseudocode for a binary search algorithm. Your algorithm should return the position of the target or -1 if not found. [6]

Q2. Explain the difference between decomposition and abstraction. Give an

example of each. [4]

Q3. The following table shows student records:

StudentID	Name	Subject	Score
1	Amy	Maths	85
2	Ben	Science	72
3	Amy	Science	91
4	Carla	Maths	68
5	Ben	Maths	74

a) Write SQL to find Amy's Science score. [2]

b) Write SQL to find the average score in Maths. [2]

c) Write SQL to list all students who scored above 80 in any subject. [2]

Q4. This Python code should find the second largest number in a list, but it contains errors:

```
def second_largest(numbers):  
    largest = max(numbers)  
    second = 0  
    for num in numbers:  
        if num > second and num != largest:  
            second = num  
    return second
```

a) What does this function return for [5, 3, 5, 2, 1]? Explain why. [2]

b) Fix the code so it works correctly for all cases. [3]

Q5. Draw the logic circuit for: $ALARM = (DOOR_OPEN \text{ OR } WINDOW_OPEN) \text{ AND NOT } ALARM_OFF$

Label all inputs and the output. [4]

Q6. Write Python code to read a CSV file "grades.csv" (format: Name,Subject,Grade) and create a new file "maths_students.txt" containing only the names of students who take Maths. [6]

Q7. Write pseudocode that asks the user for 10 numbers, validates that each is between 1 and 100, and then displays the largest valid number entered. [6]

Q8. A program stores test scores in a 2D array. `scores[row][col]` stores the score for student 'row' in test 'col'.

```
scores = [  
    [65, 70, 80],  
    [90, 85, 75],  
    [55, 60, 70]
```

]

- a) What is scores[1][2]? [1]
- b) Write Python code to find and print the average score for each student (each row). [4]
- c) Write Python code to find which student has the highest total score. Print their index (0, 1, or 2). [4]

--- ANSWER KEY (WEEK 16) ---

WORKSHEET 16A — ANSWER KEY

Q1. a) $O(n^2)$ (1)

b) Add a flag to detect if any swaps were made in a pass (1). If no swaps, the list is already sorted so the algorithm can stop early (1).

c) Pass 1: 2, 5, 1, 8 (1) — 5>2 swap, 5<8 no swap, 8>1 swap

Pass 2: 2, 1, 5, 8 (1) — 2<5 no swap, 5>1 swap, 5<8 no swap

Pass 3: 1, 2, 5, 8 (1) — 2>1 swap, 2<5 no swap, already sorted

Q2. Mark scheme (5 marks):

PROCEDURE LinearSearch(list, target) (1)

FOR i ← 1 TO LENGTH(list) DO (1)

IF list[i] = target THEN (1)

RETURN i (1)

ENDIF

NEXT i

RETURN -1 (1)

ENDPROCEDURE

Python version:

```
def linear_search(list, target):
```

```
    for i in range(len(list)):
```

```
        if list[i] == target:
```

```
            return i
```

```
    return -1
```

(Same mark allocation)

Q3. a) SELECT * FROM Products WHERE Stock < 10; (1+1)

b) INSERT INTO Products VALUES (50, 'Mouse', 25.99, 'Electronics', 100); (1+1)

Accept with field list.

c) UPDATE Products SET Price = Price * 1.1 WHERE Category = 'Electronics'; (3)

(1 for UPDATE, 1 for SET with * 1.1, 1 for WHERE)

d) SELECT Category, MAX(Price) FROM Products GROUP BY Category; (3)

(1 for MAX, 1 for GROUP BY, 1 for correct structure)

Q4. a) `mystery("Hello")` returns "HELlo" (2)

H → vowel → H, e → vowel → E, l → not → l, l → not → l, o → vowel → O

b) `mystery("Python")` returns "PythOn" (2)

P → not, y → not, t → not, h → not, O → vowel → O, n → not

c) It converts all vowels in the string to uppercase and leaves other characters unchanged (2).

Accept: "uppercases vowels in a string"

Q5. Trace table (1 mark per correct row, max 6):

Pass	num	num MOD 2	Action	count
1	27	1	num ← 26	1
2	26	0	num ← 13	2
3	13	1	num ← 12	3
4	12	0	num ← 6	4
5	6	0	num ← 3	5
6	3	1	num ← 2	6
7	2	0	num ← 1	7
8	1	1	num ← 0	8
9	0	0	exit loop	8

(Accept if student shows up to 6 rows with correct pattern. Full trace = 5 marks.)

Final output: 8 (1) [Actually counts steps until num=0]

CORRECTED: The loop exits when num = 0, after count increments. So after 27 steps, num becomes 0. Let me recalculate properly:

Pass	num	num MOD 2	Action	count
1	27	1	num ← 26	1
2	26	0	num ← 13	2
3	13	1	num ← 12	3
4	12	0	num ← 6	4
5	6	0	num ← 3	5
6	3	1	num ← 2	6
7	2	0	num ← 1	7
8	1	1	num ← 0	8

— loop exits —

Final output: 8 (1)

Q6. Mark scheme (5 marks):

`def count_words(text):` (1)

```
words = text.split(" ") (1)
```

```
return len(words) (1)
```

Accept: manual counting with loop (3 marks max for loop approach)

Alternative:

```
def count_words(text):
```

```
    count = 0
```

```
    for char in text:
```

```
        if char == " ":
```

```
            count = count + 1
```

```
    return count + 1
```

(Loop 1, condition 1, count logic 1, return 1, function def 1)

Q7. Truth table (0.5 per cell, max 7 for table + 1 for gate):

A	B	NOT A	NOT B	NOT A AND B	A AND NOT B	OUTPUT
0	0	1	1	0	0	0
0	1	1	0	1	0	1
1	0	0	1	0	1	1
1	1	0	0	0	0	0

Logic gate: XOR (1)

Q8. Mark scheme (8 marks):

```
numbers = [] (1)
```

```
FOR i ← 1 TO 5 DO (1)
```

```
    INPUT num (1)
```

```
    numbers[i] ← num
```

```
NEXT i
```

```
sum ← 0 (1)
```

```
FOR i ← 1 TO 5 DO (1)
```

```
    sum ← sum + numbers[i] (1)
```

```
NEXT i
```

```
average ← sum / 5 (1)
```

```
count ← 0 (1)
```

```
FOR i ← 1 TO 5 DO (1)
```

```
    IF numbers[i] > average THEN (1)
```

```
        count ← count + 1 (1)
```

```
    ENDIF
```

```
NEXT i
```

```
OUTPUT sum, average, count (1)
```

(Max 8 — award best 8 points)

WORKSHEET 16B — ANSWER KEY

Q1. Mark scheme (6 marks):

```
low ← 0
```

```
high ← LENGTH(list) - 1
```

```
found ← FALSE
```

```
WHILE low <= high AND found = FALSE DO
```

```
    mid ← (low + high) DIV 2
```

```
    IF list[mid] = target THEN
```

```
        found ← TRUE
```

```
        position ← mid
```

```
    ELSE IF list[mid] < target THEN
```

```
        low ← mid + 1
```

```
    ELSE
```

```
        high ← mid - 1
```

```
    ENDIF
```

```
ENDWHILE
```

```
IF found = TRUE THEN
```

```
    RETURN position
```

```
ELSE
```

```
    RETURN -1
```

```
ENDIF
```

(1 mark per significant correct step, max 6)

Q2. Decomposition: breaking a complex problem into smaller, manageable sub-problems (1). Example: making a game — split into graphics, sound, physics, scoring (1).

Abstraction: removing unnecessary detail to focus on essential information (1).

Example: using a map that shows roads but not buildings; login form hides encryption (1).

Q3. a) SELECT Score FROM table WHERE Name = 'Amy' AND Subject = 'Science'; (1+1)

b) SELECT AVG(Score) FROM table WHERE Subject = 'Maths'; (1+1)

c) SELECT * FROM table WHERE Score > 80; (1+1)

Accept with DISTINCT or name selection.

Q4. a) Returns 3 (1). The largest is 5 (at positions 0 and 2). The code finds the largest number that is not equal to 5. It scans: 5→skip (==largest), 3→second=3, 5→skip, 2→not>3, 1→not>3. So returns 3 (1).

b) Fixed code (3 marks):

```
def second_largest(numbers):  
    largest = max(numbers)  
    # Remove one occurrence of largest  
    numbers_copy = list(numbers) # don't modify original  
    numbers_copy.remove(largest)  
    return max(numbers_copy)
```

Or:

```
def second_largest(numbers):  
    unique = list(set(numbers)) # remove duplicates  
    unique.sort()  
    return unique[-2]
```

Or: track both largest and second largest in one pass.

(Any correct fix = 3 marks)

Q5. Logic circuit (1 per correct gate, 1 for connections, 1 for labels, max 4):

```
DOOR_OPEN —┐—OR—┐—AND— ALARM  
WINDOW_OPEN—┘  ┘  
ALARM_OFF —NOT—┘
```

Q6. Mark scheme (6 marks):

```
with open("grades.csv", "r") as infile: (1)  
    with open("maths_students.txt", "w") as outfile: (1)  
        for line in infile: (1)  
            data = line.strip().split(",") (1)  
            if data[1] == "Maths": (1)  
                outfile.write(data[0] + "\n") (1)
```

Accept: any working file handling approach.

Q7. Mark scheme (6 marks):

```
largest ← -1 (1)  
FOR i ← 1 TO 10 DO (1)  
    REPEAT  
        OUTPUT "Enter number"  
        INPUT num (1)  
    UNTIL num >= 1 AND num <= 100 (1)  
    IF num > largest THEN (1)  
        largest ← num (1)  
ENDIF
```

NEXT i

OUTPUT largest (1)

(Max 6)

Accept: WHILE loop for validation instead of REPEAT UNTIL.

Q8. a) 75 (1)

b) Mark scheme (4 marks):

for row in range(len(scores)): (1)

total = 0 (1)

for col in range(len(scores[0])): (1)

total = total + scores[row][col] (1)

average = total / len(scores[0]) (1)

print("Student", row, "average:", average) (1)

(Max 4)

c) Mark scheme (4 marks):

highest_total = 0 (1)

highest_index = 0 (1)

for row in range(len(scores)): (1)

total = sum(scores[row]) (1)

if total > highest_total: (1)

highest_total = total (1)

highest_index = row (1)

print(highest_index) (1)

(Max 4)

WEEK 17: MOCK EXAM 1 — PAPER 1

LESSON 17.1 — MOCK EXAM: Paper 1

Topic: Full Paper 1 mock examination

Duration: 1 hour 45 minutes (105 minutes) + reading time if applicable

Lesson Type: Formal Assessment

Syllabus Ref: All Paper 1 topics 1–6 (CIE 0478)

Learning Objective:

This is a full mock examination. Students demonstrate their understanding of all Paper 1 topics under formal exam conditions.

Instructions for Administration:

- Seating: exam arrangement (rows, spaced apart if possible)
- Materials: pen (black/blue), pencil, ruler, eraser

- NO: calculators, phones, smartwatches, notes, dictionaries (unless access arrangement)
- Reading time: 5 minutes if desired (not counted in exam time)
- Start and end times: write on board
- Students write name and date on front of paper
- If a student finishes early, they should check their work silently
- Collect all papers and question booklets at the end

EAL Support:

- Extra time as per access arrangements (discuss with SENCO)
- Bilingual word list provided if approved by exam officer
- Seating near front for clearer view of clock
- Instructions clarified before exam starts

Differentiation:

Emerging: Extra time (25% if approved); rest breaks; scribe or word processor
if in access arrangements; simplified language paper if available

Developing: Standard exam conditions

Secure: Standard exam conditions; challenge to achieve A/A* boundaries

Resources Needed:

- Printed mock exam papers (see below)
- Printed answer booklets / lined paper
- Clock visible to all students
- "Exam in progress" sign for door
- Spare pens

--- LESSON 17.2 ---

LESSON 17.2 — Mock Review & Gap Analysis

Topic: Self-marking, identifying weak areas, creating revision plan

Duration: 40 minutes

Lesson Type: Review / Feedback

Syllabus Ref: All Paper 1 topics (CIE 0478)

Learning Objective:

By the end of this lesson, students will have self-marked their mock exam, identified gaps in their knowledge, and created a personal revision plan.

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"Estimate your score out of 75 before you see the mark scheme. Write it down."

This builds metacognition — comparing estimated vs actual score.

3–30 min | SELF-MARKING

Distribute mark scheme. Students mark their own paper in a different colour.

Process:

85. Go through Section A question by question (teacher reads question + answer)

86. Students award themselves marks

87. Go through Section B

88. Go through Section C

89. Students total their marks

Teacher circulates to resolve disputes and check marking accuracy.

Grade boundaries displayed on board:

A* = 60+ marks (80%+)

A = 53–59 marks (70–79%)

B = 45–52 marks (60–69%)

C = 38–44 marks (50–59%)

D = 30–37 marks (40–49%)

E = 23–29 marks (30–39%)

U = Below 23 marks

30–38 min | GAP ANALYSIS

Students complete gap analysis template:

Topic Area	Marks Lost	Action Needed
Data Transmission		
Hardware (CPU, Memory, etc.)		
Software (OS, Translators)		
Internet & Security		
Emerging Tech & Ethics		
Databases & Boolean Logic		

Teacher collects gap analysis sheets for future planning.

38–40 min | PLENARY

"One thing you did well, one topic to revise, one question type to practise."

Students share with partner.

Homework: Begin revising your weakest topic. Use CIE textbook + online resources.

EAL Support:

- Mark scheme available with key terms in both languages
- Gap analysis template with bilingual labels
- One-to-one support for understanding marking criteria

Differentiation:

Emerging: Teacher-assisted marking; TA reads mark scheme aloud; simplified gap analysis with tick boxes; reteach of most challenging questions in small group

Developing: Self-mark independently; complete gap analysis; set revision targets

Secure: Self-mark; peer-check marking accuracy; lead discussion on tricky questions;

create additional practice questions for weak topics

Resources Needed:

- Mark scheme (printed or projected)
- Gap analysis template (printed)
- Grade boundaries poster
- Coloured pens for marking

--- MOCK EXAM PAPER 1 ---

VICTORIA PARK ACADEMY SHENZHEN

CIE IGCSE COMPUTER SCIENCE (0478)

MOCK EXAMINATION — PAPER 1

Time: 1 hour 45 minutes

Maximum Marks: 75

Name: _____

Class: _____

Date: _____

INSTRUCTIONS:

- Answer ALL questions.
- Write your answers in the spaces provided.
- Use black or blue pen. You may use pencil for diagrams.
- All questions are compulsory.
- The number of marks for each question is shown in brackets [].
- Write in full sentences where appropriate.

THIS PAPER CONTAINS THREE SECTIONS:

- Section A: Short answer questions (30 marks)
- Section B: Structured questions (30 marks)
- Section C: Extended answer (15 marks)

SECTION A: SHORT ANSWER QUESTIONS (30 marks)

Answer ALL questions.

Q1. Data can be transmitted in different modes.

- a) State the difference between serial and parallel transmission. [1]
- b) USB uses serial transmission. State one advantage of serial transmission over parallel transmission. [1]
- c) State the difference between simplex and half-duplex transmission. [1]

Q2. The Von Neumann architecture uses several registers.

- a) State the role of the Program Counter (PC). [1]
- b) State the role of the Memory Data Register (MDR). [1]

c) State the role of the Accumulator (ACC). [1]

Q3. A computer has 8 GB of RAM and a 512 GB SSD.

a) State which of these is volatile. [1]

b) Explain the purpose of the SSD. [1]

c) Give two reasons why an SSD is preferred over an HDD in a laptop. [2]

Q4. Operating systems manage computer resources.

a) State two functions of an operating system. [2]

b) What is the difference between a CLI and a GUI? [2]

Q5. Programs written in high-level languages must be translated.

a) State the difference between a compiler and an interpreter. [2]

b) State one advantage of using a compiler instead of an interpreter. [1]

Q6. The Internet and the World Wide Web are not the same thing.

Explain the difference. [2]

Q7. Cyber security is important for protecting systems.

a) Describe what a firewall does. [1]

b) State two other methods of protecting against cyber attacks. [2]

Q8. Artificial intelligence is being used in many applications.

a) Define artificial intelligence. [1]

b) Give two examples of how AI is used in everyday life. [2]

Q9. A database table called Employees has these fields:

EmployeeID, FirstName, LastName, Department, Salary

a) Write SQL to display the first and last names of all employees. [1]

b) Write SQL to display employees in the 'IT' department earning more than 50000. [2]

c) Write SQL to add a new employee with ID=25, FirstName='John', LastName='Lee', Department='Sales', Salary=45000. [2]

Q10. Complete the truth table for a NOR gate. [4]

A	B	OUTPUT
---	---	--------

0	1	
---	---	--

1	0	
---	---	--

SECTION B: STRUCTURED QUESTIONS (30 marks)

Answer ALL questions.

Q11. A company uses a local area network (LAN) to connect its computers.

a) Describe the role of a router in a network. [2]

b) Each network interface card (NIC) has a MAC address.

What is a MAC address and why is it unique? [2]

c) The company is choosing between a star topology and a bus topology.

Compare these two topologies. [4]

d) Explain the process of packet switching when an employee sends an email attachment to a colleague in another office. [4]

Q12. A student is learning about computer hardware and software.

a) Describe the Fetch-Decode-Execute cycle. Your answer should refer to the Program Counter, Memory Address Register, Memory Data Register, and Current Instruction Register. [6]

b) Cache memory is used to improve CPU performance.

Explain how cache memory improves performance. [2]

c) Describe the difference between RAM and ROM. [2]

Q13. Computer systems need to be protected from various threats.

a) Describe each of the following types of malware:

i) Virus [1]

ii) Worm [1]

iii) Trojan [1]

b) Explain how SSL/TLS provides security when a user visits a website using HTTPS. [3]

c) Two-factor authentication (2FA) is used to improve security.

Explain how 2FA works and why it is more secure than just a password. [3]

d) A school stores student personal data on a central server.

Describe three security measures the school should implement. [6]

SECTION C: EXTENDED ANSWER (15 marks)

Answer ALL parts of this question.

Q14. Automated systems and emerging technologies are changing the way businesses and individuals operate.

a) An automated greenhouse uses sensors, a microprocessor, and actuators to control the growing environment.

i) Name one sensor that could be used and explain what it measures. [2]

ii) Name one actuator that could be used and explain what it does. [2]

iii) Explain how the control system works, including the role of the sensor, microprocessor, and actuator. [4]

b) Artificial intelligence and machine learning are being used in healthcare.

i) Describe how machine learning can be used to help diagnose diseases from medical images. [3]

- ii) Discuss the ethical concerns of using AI in healthcare. [4]
- c) The increasing use of technology has environmental impacts.
Describe two negative environmental impacts of technology use and suggest how each can be reduced. [4]

END OF PAPER

--- MOCK EXAM MARK SCHEME ---

VICTORIA PARK ACADEMY SHENZHEN

CIE IGCSE COMPUTER SCIENCE (0478)

MOCK EXAMINATION — PAPER 1 — MARK SCHEME

Maximum Marks: 75

Grade Boundaries:

- A* = 60–75 marks (80%+)
- A = 53–59 marks (70–79%)
- B = 45–52 marks (60–69%)
- C = 38–44 marks (50–59%)
- D = 30–37 marks (40–49%)
- E = 23–29 marks (30–39%)
- U = 0–22 marks (0–29%)

Time: 1 hour 45 minutes

SECTION A: SHORT ANSWER QUESTIONS (30 marks)

Q1.

- a) Serial sends one bit at a time (0.5); parallel sends multiple bits simultaneously (0.5). [1]
- b) Any one: more reliable over longer distances / less interference / thinner cables / cheaper cables / no skew (1). [1]
- c) Simplex is one-way only (0.5); half-duplex is both ways but not at the same time (0.5). [1]

Q2.

- a) Holds the address of the next instruction to be fetched (1). [1]
- b) Stores data/instructions being transferred to/from memory (1). [1]
- c) Stores the result of arithmetic and logic operations (1). [1]

Q3.

- a) RAM (1). [1]
- b) Long-term / permanent storage of files and programs (1). [1]
- c) Any two: no moving parts (more durable/less likely to fail from shock) (1), faster read/write speeds (1), uses less power (longer battery life) (1),

silent operation (1), lighter (1). [2]

Q4.

- a) Any two from: memory management, file management, I/O management, interrupt handling, security, user interface, processor scheduling, error reporting, loading programs (1 each, max 2). [2]
- b) CLI = Command Line Interface, user types text commands (1);
GUI = Graphical User Interface, user clicks icons/uses windows (1). [2]

Q5.

- a) Compiler translates entire program at once into machine code (1);
interpreter translates and executes line by line (1). [2]
- b) Any one: produces standalone executable / program runs faster after compilation / all errors reported at end / source code not needed to run (1). [1]

Q6. The Internet is the global infrastructure/network of connected computers (1);

the World Wide Web is a service/system of interlinked hypertext documents accessed via browsers that runs on the Internet (1). [2]

Q7.

- a) Monitors incoming and outgoing network traffic and blocks unauthorised access based on rules (1). [1]
- b) Any two from: antivirus, 2FA, encryption, access levels, physical security, regular backups, user education, penetration testing, biometric auth (1 each, max 2). [2]

Q8.

- a) Computer systems that can perform tasks that normally require human intelligence (1). [1]
- b) Any two: voice assistants (Siri/Alexa), recommendation systems (Netflix/Amazon), medical diagnosis, fraud detection, autonomous vehicles, facial recognition, game playing (1 each, max 2). [2]

Q9.

- a) SELECT FirstName, LastName FROM Employees; (1). [1]
- b) SELECT * FROM Employees WHERE Department = 'IT' AND Salary > 50000; (1+1). [2]
- c) INSERT INTO Employees (EmployeeID, FirstName, LastName, Department, Salary) VALUES (25, 'John', 'Lee', 'Sales', 45000); (1+1). [2]
- Accept without field list.

Q10. NOR truth table (1 per correct row, max 4):

A	B	OUTPUT
0	0	1
0	1	0

| 1 | 0 | 0 |

| 1 | 1 | 0 |

[4]

SECTION B: STRUCTURED QUESTIONS (30 marks)

Q11.

- a) A router forwards data packets between networks (1). It uses routing tables to determine the best path for data to travel (1). [2]
- b) A MAC address is a unique hardware identifier assigned to a network interface card (1). It is unique because each NIC manufacturer is assigned a range of addresses, ensuring no two NICs worldwide have the same MAC (1). [2]
- c) Comparison (1 mark per valid point, max 4):
- Star: all devices connect to a central switch/hub; bus: all devices connect to a single backbone cable (1)
 - Star: easy to add/remove devices; bus: difficult to add new devices (1)
 - Star: one cable failure doesn't affect others; bus: backbone failure brings down entire network (1)
 - Star: more cabling required; bus: less cabling needed (1)
 - Star: if central switch fails, whole network down; bus: all devices share bandwidth causing collisions (1)
 - Star: generally faster as each device has dedicated connection to switch (1)
- (Max 4)

d) Packet switching process (1 mark per point, max 4):

- The email attachment is split into packets (1)
 - Each packet has a header containing source/destination addresses and a sequence number (1)
 - Packets travel independently across the network via routers (1)
 - Packets may take different routes to the destination (1)
 - Packets are reassembled at the destination in the correct order using sequence numbers (1)
 - If one route fails, packets can be rerouted, making the system fault-tolerant (1)
- (Max 4)

[Total: 12]

Q12.

a) FDE cycle (1 mark per point, max 6):

- The Program Counter holds the address of the next instruction to fetch (1)
- The PC address is copied to the Memory Address Register (1)
- The address is placed on the address bus and a read signal is sent on the control bus (1)

- The instruction is retrieved from memory and placed on the data bus, then copied to the Memory Data Register (1)
- The instruction is copied from the MDR to the Current Instruction Register (1)
- The PC is incremented to point to the next instruction (1)
- The instruction in the CIR is decoded by the control unit (1)
- The instruction is executed by the appropriate unit (ALU for calculations) (1)

(Must name at least 3 of the 4 required registers for full marks. Max 6)

b) Cache is a small, fast memory between CPU and RAM (1). It stores frequently accessed data so the CPU doesn't have to wait for slower RAM, reducing memory access time (1). [2]

c) RAM is volatile (data lost when power off), read-write, stores currently running programs (1). ROM is non-volatile (permanent), read-only (mostly), stores boot/start-up instructions (1). [2]

[Total: 10]

Q13.

a) i) A virus attaches to legitimate files/programs and needs user action to spread; it corrupts or deletes data (1). [1]

ii) A worm is a standalone program that self-replicates across networks without needing user action; it uses bandwidth and slows systems (1). [1]

iii) A Trojan disguises itself as legitimate software; it creates a backdoor allowing unauthorised access to the system (1). [1]

b) SSL/TLS security (1 mark per point, max 3):

- SSL/TLS uses encryption to secure data transmitted between browser and server (1)
- Asymmetric encryption is used during the initial handshake to establish a secure connection and exchange a session key (1)
- Symmetric encryption is then used with the session key for faster data transfer during the session (1)
- This prevents eavesdroppers from reading intercepted data (1)

(Max 3)

c) 2FA explanation (1 mark per point, max 3):

- 2FA requires two different forms of identification to verify a user (1)
- Typically: something the user knows (password) AND something the user has (phone/SMS code) or something the user is (fingerprint) (1)
- It is more secure because even if a password is stolen, the attacker still needs the second factor to gain access (1)

(Max 3)

d) Security measures (1 mark for method + 1 for description, max 6):

- Firewall: monitors and filters network traffic to block unauthorised access (1+1)
- Encryption: scrambles stored data so it cannot be read if the server is compromised (1+1)
- Access levels: different staff have different permissions — teachers can view their students' data but not all students' data (1+1)
- Regular backups: copies of data stored separately so it can be recovered if lost (ransomware, hardware failure) (1+1)
- Antivirus software: detects and removes malware that could steal data (1+1)
- Physical security: locked server room, CCTV, restricted access to server hardware (1+1)
- 2FA for all staff accounts accessing student data (1+1)
- User education: training staff to recognise phishing and follow security protocols (1+1)

(Max 6)

[Total: 14]

SECTION C: EXTENDED ANSWER (15 marks)

Q14.

a) i) Any sensor (1) + what it measures (1):

- Temperature sensor — measures air temperature (1+1)
- Moisture/humidity sensor — measures soil moisture or air humidity (1+1)
- Light sensor — measures light levels (1+1)
- pH sensor — measures soil acidity (1+1)
- CO2 sensor — measures carbon dioxide levels (1+1)

[2]

ii) Any actuator (1) + what it does (1):

- Motor/pump — opens/closes water valves for irrigation (1+1)
- Heater — warms the greenhouse when temperature is too low (1+1)
- Fan/vent — opens windows or activates cooling (1+1)
- Grow lights — provides additional light (1+1)
- Sprinkler — waters plants (1+1)

[2]

iii) Control system explanation (1 mark per point, max 4):

- The sensor continuously measures a condition (e.g., temperature) (1)
- The sensor sends data to the microprocessor (1)
- The microprocessor compares the reading to a pre-set/stored value (1)
- If the reading is outside the acceptable range, the microprocessor sends

a signal to the actuator (1)

- The actuator takes physical action (e.g., heater turns on, pump waters) (1)
- The system forms a feedback loop — the sensor continues monitoring and

the process repeats (1)

(Max 4)

[Total: 8]

b) i) Machine learning in medical imaging (1 mark per point, max 3):

- The machine learning system is trained on a large dataset of medical images

that have been labelled with diagnoses (1)

- The algorithm learns to recognise patterns and features associated with different diseases (1)

- When given a new, unlabelled image, the system can analyse it and suggest

a diagnosis based on patterns it learned during training (1)

- This can help doctors detect diseases earlier and more accurately, especially in areas with few specialists (1)

(Max 3)

ii) Ethical concerns of AI in healthcare (1 mark per point, max 4):

- Bias: if training data is not diverse, the AI may be less accurate for certain ethnic groups or demographics, leading to unequal care (1+1)

- Privacy: medical images contain sensitive patient data; storing and processing them raises privacy concerns (1+1)

- Accountability: if AI makes a wrong diagnosis, it is unclear who is responsible — the developer, hospital, or doctor (1+1)

- Job displacement: AI might reduce the need for radiologists or other specialists, causing unemployment (1+1)

- Over-reliance: doctors may become too dependent on AI and lose their own diagnostic skills (1+1)

- Consent: patients may not be aware AI is being used in their diagnosis (1+1)

(Must discuss at least 2 issues with development for full marks. Max 4)

[Total: 7]

c) Environmental impacts (1 mark for impact + 1 for reduction, max 4):

- E-waste: discarded electronics contain toxic materials; reduced by recycling schemes, longer product lifespans, WEEE directive compliance (1+1)
- Energy consumption: data centres and devices use large amounts of electricity;

reduced by renewable energy, more efficient hardware, cloud computing (1+1)

- Carbon footprint: manufacturing and transporting IT equipment produces CO₂; reduced by local manufacturing, reducing unnecessary upgrades (1+1)

- Resource depletion: rare minerals used in electronics; reduced by recycling, designing for reuse, finding alternative materials (1+1)

(Must give 2 impacts with reduction strategies for each. Max 4)

[Total: 4]

MARK SUMMARY

Section A: 30 marks

Q1: 3 Q2: 3 Q3: 4 Q4: 4 Q5: 3 Q6: 2 Q7: 3 Q8: 3

Q9: 5 Q10: 4

Section B: 30 marks

Q11: 12 Q12: 10 Q13: 14

(Adjusted to 30: Q11=12, Q12=10, Q13=8 → 30)

Actually: Q11=12, Q12=10, Q13=14 → 36. Cap Section B at 30 by scaling or by taking best 30 marks from the three questions.

RECOMMENDED: Award all marks as shown. Total may exceed 75.

The examiner should award marks as earned and then:

- If total > 75, scale proportionally
- OR: Section B has 36 marks available; take best 30 marks
- OR: adjust grade boundaries accordingly

SIMPLER APPROACH: Section B = 30 marks total

Q11 (4 parts, 12 marks) → award up to 12

Q12 (3 parts, 10 marks) → award up to 10

Q13 (4 parts, 14 marks) → cap at 8 (take best 8 marks)

Section B total = 12 + 10 + 8 = 30

Section C: 15 marks

Q14: 15 (4 parts)

GRAND TOTAL: 30 + 30 + 15 = 75 marks

GRADE BOUNDARIES APPLIED TO THIS PAPER

A* = 60–75 (80%+)

A = 53–59 (70–79%)

B = 45–52 (60–69%)

C = 38–44 (50–59%)

D = 30–37 (40–49%)

E = 23–29 (30–39%)

U = 0-22 (0-29%)

END OF MARK SCHEME

WEEK 18: MOCK EXAM 1 REVIEW & TARGET SETTING

LESSON 18.1 — Mock Review & Reteaching

Topic: Reviewing most common errors, re-teaching weak topics

Duration: 40 minutes

Lesson Type: Review / Reteach

Syllabus Ref: All Paper 1 and Paper 2 topics (CIE 0478)

Learning Objective:

By the end of this lesson, students will have reviewed the most common errors from the mock exam and received targeted re-teaching on weak topics.

Key Vocabulary:

gap analysis — 差距分析 chā jù fēn xī
common error — 常见错误 cháng jiàn cuò wù
reteach — 重新教学 chóng xīn jiào xué
misconception — 误解 wù jiě
target area — 目标领域 mù biāo lǐng yù

Lesson Structure (40 Minutes):

0-3 min | DO NOW

Display on board: "The class average for each question." (Teacher should prepare this data from marked papers.)

Students identify which topics had the lowest scores and write them in books.

"Which 3 topics do YOU need to work on most?"

3-15 min | TOP 5 COMMON ERRORS — WHOLE CLASS RETEACH

Based on mock results, re-teach the 5 most common errors. Below are the most likely candidates (teacher should adjust based on actual results):

COMMON ERROR 1: Confusing RAM vs ROM properties (2 min)

- RAM = volatile, read+write, stores running programs
- ROM = non-volatile, read-only (mostly), stores boot instructions
- Memory question template:

"RAM is ____ memory used for _____. ROM is ____ memory used for _____."

- Tip: Think R AM = Random Access Memory (temporary); R OM = Read Only Memory

(permanent)

COMMON ERROR 2: Not naming registers in FDE cycle (2 min)

- Must name: PC, MAR, MDR, CIR, ACC
- Template answer: "The PC holds _____. The address is copied to the MAR.

The _____ bus carries it to memory. The instruction is placed on the _____ bus and copied to the MDR. It is then copied to the CIR. The PC is incremented.

The instruction is decoded and executed."

- Practice: students fill in the blanks with a partner.

COMMON ERROR 3: Missing WHERE in SQL DELETE (2 min)

- DELETE FROM table WHERE condition; — without WHERE, ALL records deleted!
- Common mistake: forgetting quotes around string values
- Quick-fire SQL: teacher says task, students write query, reveal answer

COMMON ERROR 4: Explaining without developing (2 min)

- "Explain" questions need: state + explain/why/consequence
- Bad: "A firewall protects the network." (1 mark)
- Good: "A firewall monitors and filters network traffic based on rules.

It blocks unauthorised access from outside the network while allowing legitimate traffic through." (2–3 marks)

- Rule of thumb: number of marks = minimum number of distinct points

COMMON ERROR 5: Incomplete truth tables (2 min)

- Truth tables need ALL rows filled — no blanks!
- For n inputs, there are 2^n rows
- Work column by column: fill intermediate columns first
- Check: does your final output column match the gate definition?

15–35 min | STATION ROTATION / TARGETED RETEACH

Set up 3–4 stations around the room, each focused on a different weak topic.

Students rotate based on their gap analysis:

Station 1: CPU & Memory — TA-led

Activities: FDE cycle diagram completion, register function matching,
RAM vs ROM card sort

Station 2: Networks & Security — teacher-led

Activities: packet switching diagram, firewall explanation practice,
malware matching

Station 3: Boolean Logic & Databases — peer-led (Secure students)

Activities: truth table practice, SQL writing drill, logic circuit drawing

Station 4: Emerging Tech & Ethics — independent work

Activities: past paper questions, AI ethics discussion cards,
environmental impact sorting

Students spend ~5 minutes at each station (3 rotations in 15 minutes).

35–40 min | PLENARY

"Name one thing you learned or remembered today that will help you in the exam."

Each student shares with class or writes on exit ticket.

Homework: Use the revision plan template to plan your revision for Y10S2.

Focus on your TOP 3 weak topics.

EAL Support:

- Bilingual key terms for all retaught topics
- Visual aids and diagrams at each station
- Sentence starters for explanation questions
- TA support at the most language-dependent station

Differentiation:

Emerging: Attend only 2 stations with TA support; simplified activities;

pre-completed notes to take away; visual learning materials

Developing: Attend 3 stations; complete all activities; identify own weak areas;

set specific revision goals

Secure: Lead Station 3 as peer tutors; attend 3–4 stations; create additional

practice questions for weak areas; help evaluate quality of peer explanations

Resources Needed:

- Mock exam score data (displayed)
- Station materials (prepared in advance)
- FDE cycle diagrams (blanks + completed)
- SQL quick-fire cards
- Truth table practice sheets
- Exit tickets
- Gap analysis results

--- LESSON 18.2 ---

LESSON 18.2 — Individual Target Setting & Next Steps

Topic: Creating personal revision plans, setting targets, previewing Y10S2

Duration: 40 minutes

Lesson Type: Reflection / Planning

Syllabus Ref: N/A (review and transition)

Learning Objective:

By the end of this lesson, each student will have created a personal revision plan with three specific targets and will understand the structure of Y10S2.

Key Vocabulary:

target setting — 目标设定 mù biāo shè dìng

revision plan — 复习计划 fù xí jì huà

SMART target — SMART 目标 SMART mù biāo

self-reflection — 自我反思 zì wǒ fǎn sī

next steps — 下一步 xià yī bù

Lesson Structure (40 Minutes):

0–3 min | DO NOW

"Write down your mock exam grade and the three topics you lost the most marks on."

Students refer to their gap analysis from Lesson 18.1.

3–10 min | SMART TARGETS

Explain SMART targets:

- S = Specific (not "do better" but "practise SQL queries 3 times per week")
- M = Measurable (you can check if you've done it)
- A = Achievable (realistic for your current level)
- R = Relevant (directly addresses a weak area)
- T = Time-bound (done by a specific date)

Example targets for different levels:

Emerging student:

- 90. "I will memorise the 5 CPU registers and their functions by Friday."
- 91. "I will complete 10 SQL practice questions with 80% accuracy by next week."
- 92. "I will draw and label the FDE cycle from memory by the end of this week."

Developing student:

- 93. "I will practise explaining 'why' in 6-mark questions, completing 5 per week."
- 94. "I will achieve 70% on a hardware topic test by next Friday."
- 95. "I will revise all 6 logic gates and complete truth tables for each."

Secure student:

- 96. "I will complete a full Paper 1 past paper under timed conditions, aiming for 65+."
- 97. "I will teach one revision session to a peer on their weakest topic."
- 98. "I will complete the CIE-specified programming project outline."

10–25 min | INDIVIDUAL TARGET SETTING

Students complete WORKSHEET 18A (Target Setting Sheet) and WORKSHEET 18B (Revision Plan Template).

Teacher circulates and discusses targets with each student individually.

Check that targets are SMART and appropriate.

Questions to ask each student:

- "What was your strongest topic? How will you maintain it?"
- "What was your weakest topic? What will you do to improve it?"
- "How much time can you realistically spend on revision each week?"
- "What resources will you use? (textbook, online, past papers, peer study)"
- "Who can help you? (teacher, TA, parent, peer)"

25–35 min | INTRODUCTION TO Y10S2

Brief overview of what comes next:

Y10S2 OVERVIEW:

- Weeks 1–6: Paper 1 topics in depth (Data Representation, Data Transmission, Hardware, Software, Internet, Security, Emerging Tech)
- Weeks 7–10: Paper 2 topics in depth (Algorithms, Programming, Databases, Boolean Logic)
- Weeks 11–12: Full mock exam preparation
- Week 13: Mock Exam 2 (Paper 1 + Paper 2)
- Weeks 14–18: Targeted revision based on mock results

What students need to do over the break:

- Complete the revision plan
- Use recommended online resources:
 - CIE past papers: <https://papers.gceguide.com>
 - SQL practice: <https://sqlbolt.com>
 - Python practice: <https://www.w3schools.com/python>
 - Boolean logic: Logicly (circuit simulator)
- Keep a revision log
- Bring completed revision to first lesson back

35–40 min | PLENARY

Each student shares ONE target with the class or their partner.

Teacher collects target sheets for follow-up.

"What is ONE thing you will do differently in Y10S2?"

Final words: "Your mock result is not your final grade. It tells you where to focus. With consistent effort, every student in this room can improve. I believe in you."

Homework: Complete the revision plan. Begin with Target 1.

EAL Support:

- Target setting sheet in both languages
- Sentence starters for targets: "I will... by... because..."
- Visual Y10S2 timeline/roadmap
- Parent communication letter (bilingual) explaining how to support revision
- Audio recording of Y10S2 overview for listening practice

Differentiation:

Emerging: Teacher-assisted target setting; pre-written targets to select from; simplified revision plan; paired with buddy for accountability; weekly check-ins

Developing: Write own targets with teacher guidance; complete revision plan; set weekly revision schedule; monthly check-ins

Secure: Independent target setting; create own revision resources; peer mentor for emerging students; lead revision groups; access advanced materials early

Resources Needed:

- WORKSHEET 18A (Target Setting Sheet)

- WORKSHEET 18B (Revision Plan Template)
- Y10S2 overview slide
- Recommended resources list (printed)
- Revision log template
- Pens and highlighters
- Parent letter (optional)

--- WORKSHEET 18A ---

WORKSHEET 18A: Mock Exam Target Setting

Student Name: _____ Date: _____

PART 1: MY MOCK RESULTS

My score: _____ / 75

My percentage: _____%

My estimated grade: _____

Topic breakdown (from your gap analysis):

Topic Area	Marks Available	Marks Scored	% Score
Data Transmission			
Hardware (CPU, Memory, etc.)			
Software (OS, Translators)			
Internet & Security			
Emerging Tech & Ethics			
Databases & Boolean Logic			

My TOP 3 strong topics:

99. _____

100. _____

101. _____

My TOP 3 weak topics:

102. _____

103. _____

104. _____

PART 2: MY THREE TARGETS

Target 1 (focus on weakest topic):

How I will achieve this:

By when: _____

How I will measure success:

Target 2 (exam technique / skill):

How I will achieve this:

By when: _____

How I will measure success:

Target 3 (maintaining strengths):

How I will achieve this:

By when: _____

How I will measure success:

PART 3: MY REVISION RESOURCES

I will use these resources:

☐ CIE textbook (chapters: _____)

☐ Past papers (source: _____)

☐ Online resources (list: _____)

☐ Revision notes (making my own: Y / N)

☐ Flashcards (making my own: Y / N)

☐ Peer study group (members: _____)

☐ Teacher/TA support (topics: _____)

☐ Parent/family support (how: _____)

Time I can spend per week on CS revision: _____ hours

Best days/times for me to revise: _____

Student signature: _____

Teacher signature: _____

Parent signature: _____ (optional)

--- WORKSHEET 18B ---

WORKSHEET 18B: Semester 2 Revision Plan Template

Student Name: _____ Date: _____

WEEK-BY-WEEK REVISION SCHEDULE

Complete this plan for the first 6 weeks of Y10S2.

WEEK 1:

Topics to revise: _____

Resources: _____

Activities: _____

Target test score: _____%

WEEK 2:

Topics to revise: _____

Resources: _____

Activities: _____

Target test score: _____%

WEEK 3:

Topics to revise: _____

Resources: _____

Activities: _____

Target test score: _____%

WEEK 4:

Topics to revise: _____

Resources: _____

Activities: _____

Target test score: _____%

WEEK 5:

Topics to revise: _____

Resources: _____

Activities: _____

Target test score: _____%

WEEK 6:

Topics to revise: _____

Resources: _____

Activities: _____

Target test score: _____%

DAILY REVISION LOG (copy this for each revision session)

Date: _____ **Day:** _____

Topics covered: _____

Time spent: _____ minutes

Resources used: _____

What I learned: _____

What I still need to work on: _____

Self-rating: 1 2 3 4 5

(circle one: 1 = not helpful, 5 = very helpful)

--- ANSWER KEY (WEEK 18) ---

WORKSHEET 18A & 18B — TEACHER NOTES

These worksheets are personalised planning tools — there are no "right answers."

Use these marking/feedback guidelines:

WORKSHEET 18A — Assessment Criteria:

- All sections completed: 1 mark
- Mock results accurately recorded: 1 mark
- 3 targets written: 1 mark (all 3 present)
- Targets are SMART (check each one):
 - Specific: names a concrete action (1)
 - Measurable: includes how to check progress (1)
 - Achievable: realistic for this student (1)
 - Relevant: addresses actual weak areas from mock (1)
 - Time-bound: has a deadline (1)
- Revision resources listed: 1 mark
- Time commitment realistic: 1 mark

(Max: check for completeness and quality of targets)

WORKSHEET 18B — Assessment Criteria:

- All 6 weeks planned: 1 mark
- Topics specific (not vague "revise CS"): 1 mark

- Resources identified: 1 mark
- Activities are varied and active (not just "read notes"): 1 mark
- Daily revision log template completed for at least one session: 1 mark

(Max: check for completeness and specificity)

FEEDBACK TO GIVE:

"Your targets are strong because..."

"To make Target ___ stronger, you could..."

"I suggest adding ___ to your revision plan."

"Your plan shows good understanding of your weak areas."

"Consider asking ___ for help with ___ topic."

FOLLOW-UP ACTIONS FOR TEACHER:

- ☐ Photocopy targets for your records
- ☐ Share relevant targets with TA for support planning
- ☐ Group students by weak topic for targeted intervention
- ☐ Schedule check-in meetings (Week 2, Week 4 of Y10S2)
- ☐ Share targets with parents (optional, with consent)
- ☐ Prepare differentiated resources for first Y10S2 topics